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Vegetation gradients in palsa – thermokarst permafrost features in Finnmark, Norway

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Declaration of Originality

Hereby, I, Anna Vollmer (matriculation number 1883880), declare that I have written the present thesis, titled

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Abstract

The rise in global average temperatures of 1.1°C from 2011 to 2020 since the pre-industrial reference period (1850–1900) has led to significant changes in arctic vegetation. This applies particularly to palsa structures in arctic peatlands, which collapse once the permafrost thaws, creating new conditions that influence the composition of the plant community. A frequently striking change in vegetation is the spread of shrubs in the Arctic which can affect the thawing process and temperature regime in palsas. The aim of this thesis was to investigate vegetation composition and cover along a gradient from intact to deeply thawed palsas in Finnmark, northern Norway and how vegetation affects permafrost thaw was investigated in a thermokarst. In addition, plant species composition and the specific role of sedges on methane emissions was analysed in a thermokarst. In intact palsas, active layer depth beneath *Betula nana* was 10 cm less in the late growing season than beneath low vegetation patches, indicating that shrubs could slow down the thawing of permafrost. Along the palsa slopes from intact palsa plateau (mean active layer depth 64 cm) to the slope base (active layer depth mostly >160 cm; always >85 cm), slope sections and active layer depth explained 19% of the difference in plant species composition. Total plant cover, cover of sedges, of *Empetrum nigrum* and *Rubus chamaemorus* changed and contributed mainly to the differences in species composition along the palsa slopes. In a thermokarst, vegetation zones differed in species composition at relatively short distance. Total biomass was considerably lower in one of the three sampled vegetation zones, which consisted of sedges only. In this zone sedge roots had smaller diameter compared to another vegetation zone. Methane emissions were three times higher in sedge compared to non-sedge patches towards the end of the growing season. The results from this work show that vegetation varies with permafrost degradation state and indicate, that an increase in *B. nana* can hamper permafrost thaw. High abundance of sedges after thaw can feed back to climate warming through increasing net emissions of methane and increasing uptake of carbon dioxide and peat accumulation. The net carbon balance however remains unclear and needs further investigation.

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1. Introduction

1.1 Effects of climate change on vegetation

Global average surface temperature increased by 1.1°C from 2011 to 2020 compared to the preindustrial reference period from 1850 to 1900 through the increase of greenhouse gas emissions such as carbon dioxide and methane in the atmosphere through anthropogenic activity (Calvin et al. 2023). In the Arctic, warming was four times stronger than average global warming for the period from 1979 to 2021, referred to as arctic amplification (Rantanen et al. 2022). The Arctic can be defined by latitude as the area north of the arctic circle (Federov et al. 2019). The accelerated warming in the Arctic among others relates to feedbacks from lowered albedo through reduction in snow and ice cover and increased water vapor affecting heat fluxes (Mark C. Serreze and Barry 2011; Vihma et al. 2016). The extent of warming varies among arctic regions and seasons, being strongest in winter (Serreze et al. 2026; Matthes et al. 2015). Heat waves in summer have become more frequent and warmer, increasing probability of warm extended periods such as in summer 2024 in northernmost Finnland with 25 days with temperatures above 25°C (Rantanen et al. 2025). Warming increases evapotranspiration and was related to an overall increase in annual precipitation across the arctic (Bintanja and Selten 2014; Box et al. 2019; Serreze et al. 2025). Precipitation increase was especially strong in the cold season (October to May) and more frequently caused rainfall in winter in relatively warm arctic climates such as northern Fennoscandia (Irannezhad et al. 2016; Box et al. 2019). Temperatures above zero in winter and autumn become more frequent and affect snow properties (Hansen et al. 2014; Barrere et al. 2018). These changes in climate affect ecosystems and cause widespread vegetation changes (Prager et al. 2020; Mekonnen et al. 2021). In wide parts of the Arctic, plant biomass has been increasing, which was mainly related to increasing summer air and soil temperature and permafrost thaw (Berner et al. 2020; Mekonnen et al. 2021). Especially deciduous shrubs, including *B. nana* have been found to increase in growth and size and expand northward and upslope (Elmendorf et al. 2012; Myers-Smith et al. 2011; Mekonnen et al. 2021).

1.2 Peatland permafrost

Approximately 20% of the land surface of the northern hemisphere is underlain by permafrost, which occurs in mountain regions and in the Arctic (Obu et al. 2019). Permafrost is defined as rock or ground, that is frozen for at least two consecutive years (Dobinski 2011). In the discontinuous permafrost zone, at the border of the permafrost zone, permafrost appears in

peatlands (Seppälä 2011). Thin snow layers allow frost formation in deep peat. For example, under 20 cm of snow, soil frost reach down to 70 cm depth in northern Finnland (Seppälä 1990). Ice lenses form, and due to increased volume after freezing, peat mounds with ice cores rise 0.5 - 10 m above the mire surface to so called palsas (Seppälä 2011). In northern Norway, which lies within the discontinuous permafrost zone, peatland permafrost formed 950 – 100 calibrated years before present (cal yr BP) (Kjellman et al. 2018).

The formation of permafrost affects vegetation as rising palsas get more drained and more exposed to winds and colder temperatures (Zuidhoff and Kolstrup 2005). With ongoing climate change however, permafrost thaw has been accelerated (Jorgenson et al. 2006; Borge et al. 2017; Chen et al. 2021a). In Finnmark, the northernmost county of Norway, 33 - 71% of palsa structures disappeared from 1950 - 2010 (Borge et al. 2017). Important factors of permafrost degradation are high summer temperatures and precipitation, and additionally wildfires or erosion events (Jorgenson et al. 2006; Hu et al. 2010; Jones et al. 2015). Dry organic soils and vegetation such as thick moss layers can on the other hand effectively isolate ground and reduce the thaw of permafrost (Fisher et al. 2016; Yi et al. 2007).

As shrubs increase in arctic regions (Mekonnen et al. 2021; Myers-Smith et al. 2011; Elmendorf et al. 2012) and capture insulative snow during winter (Sturm et al. 2001), the question has been risen, if and how shrubs affect permafrost. Several arguments speak for shrubs affecting permafrost degradation positively or negatively. Thin snow cover is important for permafrost formation and preservation (Seppälä 2011) while thick snow cover has been found to accelerate permafrost thaw due to isolation preventing complete freezing of the active layer in winter (Johansson et al. 2013). The active layer is the uppermost layer thawing annually during the growing season. Even though there are arguments for deeper active layers beneath shrubs, several studies showed that shrubs such as *B. nana* can hamper active layer deepening (Blok et al. 2010; Cameron et al. 2024; Yi et al. 2007). Both lower soil temperatures and soil moisture underneath shrubs reduced active layer deepening. Lower temperatures beneath shrubs were related to shadow and higher percentage of latent heat flux. Lower soil temperatures were attributed to water uptake by the plants and increased transpiration (Cameron et al. 2024; Aguirre et al. 2021; Blok et al. 2010; Fisher et al. 2016). Contrastingly, shrubs were also found to lead to deeper active layers due to lower albedo of branches protruding from the snow in spring (Wilcox et al. 2019). The influence of shrubs on active layer depth could potentially vary from region to region and from year to year, depending on factors such as soil, seasonal distribution and amount of precipitation and magnitude of warm temperatures.

1.3 Vegetation on degrading peatland permafrost

The thawing and degradation of permafrost soils can impact vegetation (Zuidhoff and Kolstrup 2005; Lantz et al. 2009; Magnússon et al. 2020). When the permafrost in palsas thaws, peat subsidence initially occurs at the edges and is accompanied by a significant deepening of the active layer (Seppälä 2011; Schuur et al. 2015; Martin et al. 2021). As the palsa continues to collapse, thermokarst features form, which lie several meters lower than the previous palsa plateau. The degradation of palsas can disturb the physical peat structure, disturb plant cover and form unvegetated areas. Further consequences are changing hydrological pathways, commonly leading to accumulation of water in thermokarsts (Schuur et al. 2015; Olefeldt et al. 2021). Along palsa slopes from intact palsa plateau to palsa base, vegetation may respond to changes in wind exposure, temperatures, snow depths, peat moisture (Zuidhoff and Kolstrup 2005) and to active layer deepening. Active layer deepening was suggested to favor deep rooting species partially due to increased soil volume and nitrogen release from the newly thawed permafrost, close to the thaw front (Blume-Werry et al. 2019; Keuper et al. 2017). However this may also be due to adaptations to wet conditions that can appear at the water logging thaw front (Vartapetian and Jackson 1997; Taylor 1971; Tyler et al. 2021; Blume-Werry et al. 2019). During palsa degradation and ground subsidence, peat surface gets more irregular and cracks form, increasing the number of ecological niches and plant diversity (Zuidhoff and Kolstrup 2005). Thus, degraded areas of palsas at the transition to thermokarst could show higher plant diversity, compared to intact palsa tops. Similar to species composition there are few studies on plant diversity along palsa slopes (Zuidhoff and Kolstrup 2005).

1.4 Vegetation in thermokarsts

The transition from palsa to thermokarst features has been described to cause vegetation shifts, typically to mire vegetation when the water level is high. Peat mosses, *Sphagnum spp.* and sedges (Cyperaceae) establish and new peat accumulates (Olefeldt et al. 2021; Camill 1999b, 1999a; Heffernan et al. 2022). The genus *Sphagnum* plays an important role in mire and peat formation (Dierßen 1996). *Sphagnum* keeps growing at the top of accumulating and thus rising peat and releases acids. Thus, peatland succession is associated with a decrease in pH and moisture (Rydin and Jeglum 2013). Accordingly, vegetation varies along water chemistry, pH and moisture gradients (Camill 1999b, 1999a; Jabłońska et al. 2025; Rydin and Jeglum 2013). Also, sedges have been suggested to be like *Sphagnum spp.* ecosystem engineers for their ability to create conditions favoring their own growth (Palozzi and Lindo 2017). Mire succession in thermokarsts from aquatic fen to more dry hummocky bogs was described in several studies

(Heffernan et al. 2022; Gałka et al. 2017; Camill 1999b, 1999a). Successive peat subsidence leads to different time spans since thaw and may thus relate to varying plant successional stages within thermokarst. Varying amounts of peat subsidence may also lead to moisture differences, for example at the edge versus the center of the thermokarst and can affect plants establishing on the disturbed peat. Therefore, water table depth can vary both in time and space within thermokarst and may lead to small scale variation in vegetation. *Sphagnum riparium* was commonly found together with a high abundance of sedges early after thaw, when the water table is high and conditions are particularly wet (Heffernan et al. 2022; Camill 1999b). As peat rises further up the water table *S. angustifolium* and later *S. fuscum* may dominate (Gałka et al. 2017; Camill 1999b, 1999a; Horton et al. 1979). However, species composition varies within thermokarsts, especially along chemical and hydrological gradients, when thermokarsts are not entirely isolated (Jabłońska et al. 2025).

1.5 Methane emissions from thermokarsts

In anaerobic zones of vegetated thermokarsts, decomposition of plant detritus can lead to microbial methane production. There are two main methane production pathways, one by reducing CO_2 to CH_4 and oxidizing H_2 (hydrogenotrophic methanogenesis) and the other by splitting acetate and reducing the methyl group (acetoclastic methanogenesis) (Ferry 2011). Acetate and CO_2 plus H_2 result from fermentation of peat and relatively fresh plant litter. Methane can get oxidized to CO_2 in small amounts in the anaerobic layer, but mostly when methane diffuses to aerobic zones (Ferry 2011; Laanbroek 2010). Sedges can affect methane emissions in two ways. First, they have enlarged intercellular, connected cells, so-called aerenchyma, in stems and roots allowing bidirectional exchange of gases between the atmosphere-leaf interface and the root-peat interface (Vartapetian and Jackson 1997). The passive diffusive transport of gases follows the concentration gradient between atmosphere and root-peat interface (Ge et al. 2024). Particularly oxygen is transported to the root system where it is mainly consumed by root respiration allowing growth in otherwise anaerobic conditions (Vartapetian and Jackson 1997). A part of the oxygen is lost radially around the roots, creating aerobic conditions in the rhizosphere (Vartapetian and Jackson 1997; Haviland and Noyce 2024). Under aerobic conditions in the rhizosphere, methane can be partly oxidized by methanotrophic bacteria which can significantly reduce the emission into the atmosphere (Agethen et al. 2018; Haviland and Noyce 2024). A large fraction of methane is transported via aerenchymatous cells into the atmosphere following the concentration gradient (Jentzsch et al. 2024; Magnússon et al. 2020). The second effect of sedges is the production of aboveground

and belowground litter that is fermented under anoxic conditions, providing intermediates such as acetate, hydrogen and carbon dioxide (Ferry 2011; Ström et al. 2012; Mollenkopf et al. 2026). Several studies found an increase of methane emissions with an increase of sedges (Ström et al. 2012; Jentzsch et al. 2024; Mollenkopf et al. 2026; Ding et al. 2005). Thus, accelerated thermokarst formation with often sedge favouring conditions (Chen et al. 2021; Camill 1999b; Gałka et al. 2017; Heffernan et al. 2022), increases the potential for methane emissions.

1.6 Aim and hypotheses

The aim of this study is to evaluate vegetation gradients on permafrost degradation gradients related to ongoing climate change. The work is structured in three parts according to degradation status of permafrost with analyses of 1) intact palsas, 2) palsa slopes from intact to degraded palsas at the transition to thermokarst and 3) a thermokarst. Since *B. nana* increases with climate warming and was found to affect active layer depth, here active layer depth is measured on intact palsas beneath *B. nana* and adjacent low vegetation. Vegetation on gradients from intact to degraded palsas have received little attention. Here species composition, including abundance of the occurring species, in relation to active layer depth and palsa slope section, and corresponding changes in plant diversity along the palsa slopes are investigated. In a thermokarst species composition, biomass, root morphology of sedges and effects of sedges on methane emissions are analyzed. The following is hypothesized.

1 Palsa

1.1 Active layer is shallower under relatively high dwarf shrub *Betula nana* compared to patches with low vegetation.

2 Plant gradient along palsa slopes

2.1 Plant cover and plant species composition change within palsa slopes from palsa top, upper slope, lower slope to slope base. The change in species composition in these sections follows the pattern in active layer depth.

2.2 Plant species diversity is highest at the base of palsas, and lowest at the palsa top.

3 Vegetation composition in a thermokarst and impact of sedges on methane fluxes

3.1 Species composition, living biomass and root morphology vary within thermokarst and are related to the successional stage.

3.2 Methane emissions are higher in patches with fully grown sedges compared to where sedges have been removed.

2. Methods

2.1 Study sites

The study was conducted in permafrost peatlands in Finnmark, northern Norway. Permafrost aggregated in peatlands around 950 to 100 calibrated years before present (cal yr BP) in Finnmark (Kjellman et al. 2018) and is currently thawing due to climate warming (Jorgenson et al. 2006). At the study sites there are palsa and thermokarst formations with varying palsa and thermokarst sizes. Vegetation is sporadic, generally low and consists mainly of dwarf shrubs, bryophytes and lichen (*Cladonia spp.*). Common species are *Empetrum nigrum, Betula nana, Rhododendron tomentosum, Vaccinium uliginosum, Vaccinium vitis-idea and Rubus chamaemorus*. Thaw depressions often fill with water and appear as thermokarst mires with mainly *Sphagnum spp.* and sedges (*Eriophorum and Carex)*. On palsa plateaus vascular vegetation is lower than in more protected depressions and slopes. There were three study sites, Ádejávri, Iškoras and Karlebotn. Distance to the respectively closer study site is approximately 200 km.

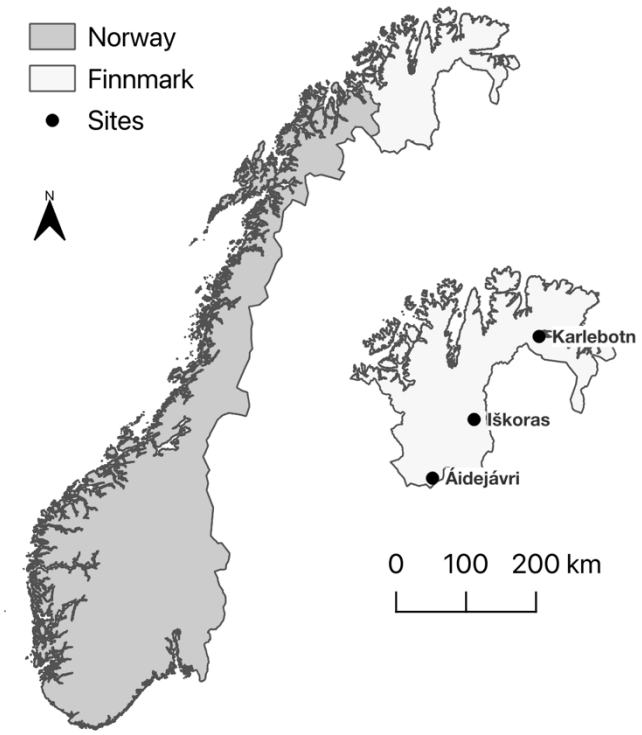


Figure 1: Map of the study sites in Finnmark, northern Norway. For this work the main study site was Ádejávri.

2.1.1 Ádejávri

Ádejávri site is located approximately 68.7 North and 23.3 East and ca. 373 m above mean sea level (a.m.s.l.). Palsa heights and thaw depression depths vary in the range of a few meters depending on the extent of permafrost degradation. Active layer depth at the end of the growing season was up to ~60 cm deep on palsa plateaus. Depressions typically result in thermokarst mires with dominating *Sphagnum spp.* while less degraded depressions sometimes show a high abundance of lichen (*Cladonia spp.*). Commonly thermokarst ponds are surrounded by intact palsas and are thus hydrologically isolated (Figure 2). Mean annual air temperature was -1.6°C for the reference period 1991 to 2020 and -3.1°C for 1961 to 1990 measured at Šihččajávri weather station, approximately 9 km from Ádejávri (382 m above mean sea level, station number SN93900). The annual precipitation was 570 mm for the reference period from 1991 to 2020 and 360 mm from 1961 - 1990 measured at the same weather station using data from the Norwegian Meteorological Institute.



Figure 2: Example of A) a palsa and B) a thermokarst at Ádejávri site.

2.1.2 Iškoras

Iškoras study site is located approximately 69.3 North and 25.3 East, ~ 380 m a.m.s.l. next to a ~640 m high peak (Iškorasfjellet) (Lehmann et al. 2023). Fens formed and accumulated peat approximately 9200 cal yr BP. They formed to permafrost peat plateaus at the beginning of the little ice age ~800 cal yr BP (Kjellman et al. 2018). At this site permafrost is in comparison more degraded and palsas are fewer and smaller than at Ádejávri site. They are surrounded by large fens (Kjellman et al. 2018) and the Báhkiljohka catchment (Lehmann et al. 2023). Thus, thermokarst ponds are hydrologically less isolated than at Ádejávri site. In thermokarsts and

fens *Sphagnum spp., Eriophorum spp.* and *Carex spp.* dominate. Mean annual temperature was -2.4°C for the reference period from 1961 to 1990 and -1.2°C from 1991 to 2020 measured at Karasjok-Markannjarga weather station (SN97251) 131 m a.m.s.l. approximately 20 km from Iškoras. Annual precipitation was 370 mm for the reference period from 1961 to 1990 and 420 mm from 1991 to 2020 measured at the same station using data from the Norwegian Meteorological Institute.

2.1.3 Karlebotn

Karlebotn is located approximately 70.1 North and 28.6 East, at the coast and 26 m a.m.s.l. (Kjellman et al. 2018). The peatland developed ~5150 cal yr BP, later than at more continental sites such as Iškoras probably because of glaciation and uplift. Permafrost aggregated ~950 cal yr BP (Kjellman et al. 2018). Karlebotn was the largest study site with extensive mire areas, and areas with well-connected relatively tall palsas and isolated thermokarsts. According to the weather station Bergebydalen (SN98974, 150 m a.m.s.l., 20 km from the study site) mean annual air temperature was 0.8°C and annual precipitation was 515 mm in 2025 using data from the Norwegian Meteorological Institute.

2.2 Selection of survey and sampling locations

Data for all three hypotheses was collected at the main study site Ádejávri and additionally at Iškoras and Karlebotn for Hypothesis 2. For analysing the effects of *Betula nana* on active layer depth (Hypothesis 1), *Betula nana* patches with adjacent patches of low vegetation (mainly *Empetrum nigrum, Rubus chamaemorus, Vaccinium vitis-idea and* bryophytes) on palsa plateaus were selected across Ádejávri study site. To analyse the transition of plants from palsa top to slope base (Hypothesis 2), plant cover and composition were determined at three to four palsa slopes at each study site. Palsa slopes differed in exposition, degree of incline, total plant cover, soil disturbance (e.g. cracks) and length, covering a wide range of palsa slopes.

At Ádejávri, a thermokarst mire was selected that had visually different vegetation zones (Figure 3). For analysing variability in species composition, seven visually different areas of the thermokarst were chosen for sampling (AT1-7; Table 1). Additionally, five biomass samples were taken within the vegetation zones from AT1, AT3 and AT5 each. AT1, AT3 and AT5 correspond to recent, young and mature from the work of Arwen Henz.

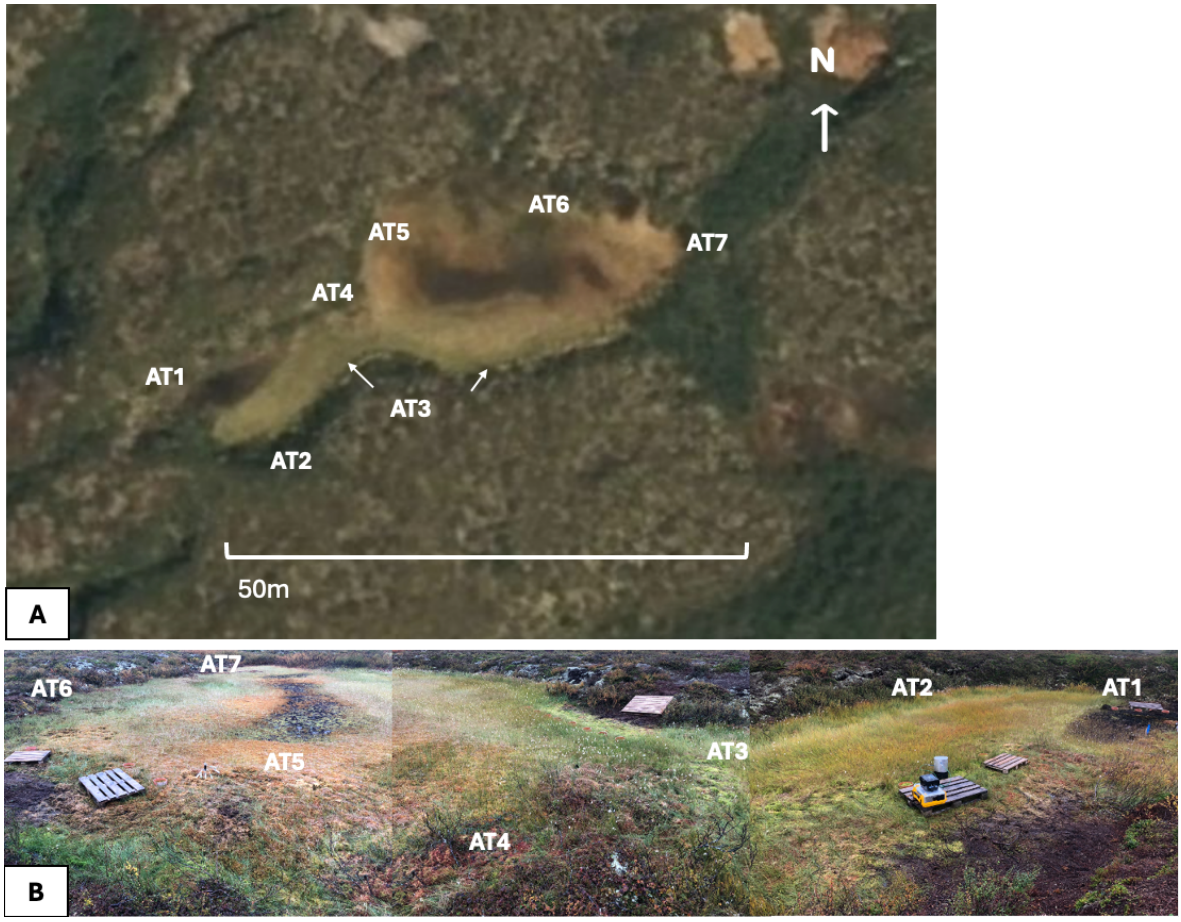


Figure 3: A) Aerial photo and B) closer photos with an overview of sampling locations AT1-7 in the selected thermokarst at Ádejávri site. The screenshot of the aerial photo was obtained from ©norgeskart.no.

Table 1: Description of the vegetation zones at Ádejávri thermokarst from which samples were taken.

Vegetation zone	**Description**
AT1	Bare peat, sporadic abundance of sedges
AT2	Wet thermokarst margin with floating *Sphagnum spp.*
AT3	Wet thermokarst with dense cover of light green *Sphagnum spp.*, high abundance of a sedge species
AT4	Moist thermokarst with partially red *Sphagnum spp.* and dwarf shrubs
AT5	Wet thermokarst with dark orange *Sphagnum spp.*; lower abundance of sedges
AT6	Moist thermokarst edge with dark orange and red *Sphagnum spp.*, sporadic abundance of dwarf shrubs and sedges
AT7	Less moist, higher variety in colour, and more dense *Sphagnum spp.*, appearance of dwarf shrubs

2.3 Plant survey and sampling

2.3.1 Palsa

To analyse the effect of *Betula nana* on active layer depth, cover of species, active layer depth and vegetation height were recorded in 0.5 x 0.5 m plots at Ádejávri site. For each location there was a *Betula nana* and a low vegetation plot selected with 0.5 m to 2 m distance, so that plots were paired. With paired plots other effects on active layer depth could be minimized. Active layer depth and vegetation height were measured three times in each plot. Active layer depth was measured with a thaw probe that reached up to 1.6 m depth. Vegetation height was measured at the tall species occurring which were *B. nana* and *Rhododendron tomentosum* in *B. nana* patches and mainly *Empetrum nigrum* in low vegetation. Vegetation cover was visually estimated as percent cover of each species. Lichen and bryophytes species were summarized respectively. The difference in *B. nana cover* was the criterion for choosing the plots and necessary for answering the hypothesis.

2.3.2 Plant gradients along palsa slopes

The surveys along the palsa slope always extended from the palsa top down the slope to the slope base, at the transition from palsa to thermokarst. Surveys were done in 0.5 m x 0.5 m plots with ca. 1.5 m distance to each other to estimate cover of species for vascular plants and of groups lichen and bryophytes for non-vascular plants. If identification was not possible in the field, plants were sampled for later identification, which was the case for sedges. In each plot, active layer depth (up to 1.6 m) and soil moisture (µS) were measured once with three repetitions and soil temperature with two repetitions. Soil moisture was recorded at 5 cm depth using a TDR probe (ML2x ThetaProbe; Delta-t, London, United Kingdom). Soil temperature was obtained at 10 cm depth after three minutes, so that the probe could adjust to the ground temperature before measuring. Exposition was noted at each palsa slope.

2.3.3 Thermokarst

At all seven thermokarst sampling locations, respectively, three samples were taken to identify species. At sampling locations AT1, AT3 and AT5, adjacent to gas flux measurements of the work of Arwen Henz, five samples were additionally taken to determine biomass and sedge

root morphological parameters (length, diameter and surface area). Samples were taken with a cylindrical tube with a diameter of 9.4 cm and a height of 35 cm, so that sample sizes were similar and biomass samples had the same volume with a sample depth of 30 cm.

To analyse the influence of sedges on methane fluxes, there have been 30 cm deep (in total 35 cm) cylindric (diameter 23.5 cm) collars installed at vegetation zone AT3 (Figure 3). In four out of eight collars sedges have been removed right under the surface, so that there were four sedge and four non-sedge collars with otherwise similar conditions. A chamber was placed and sealed airtight on top of the collars. To measure methane fluxes the chamber was connected to a portable greenhouse gas analyser (UGGA; Los Gatos Research, San Jose, CA, USA), which pumped gas from the chamber through the UGGA for approximately five minutes to obtain a linear flux rate (ppm s^{-1}). The value of the slope of the flux rate was fitted and calculated afterwards with linear regression and used for calculating the methane flux (in mg m^{-2} d^{-1}). Fluxes were obtained and calculated for most of the growing season by Lena Kaiser.

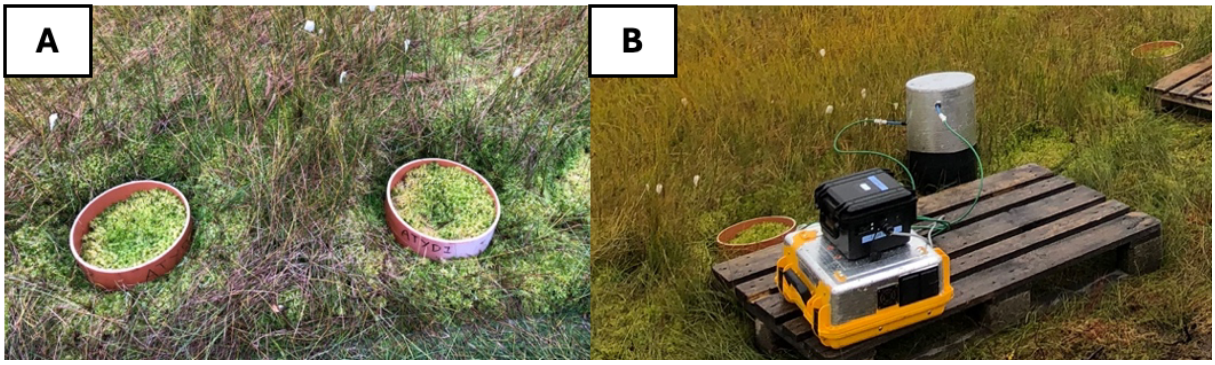


Figure 4: Non-sedge collars with removed sedges (A) and portable greenhouse gas analyser (UGGA; Los Gatos Research, San Jose, CA, USA) and chamber that is placed on a collar for gas flux measurements (B) in thermokarst vegetational zone AT3 at Ádejávri.

2.4 Sample and data processing

2.4.1 Plant gradient along palsa slopes

Each individual palsa slope was divided into four consecutive slope sections: ‘palsa top’, ‘upper slope’, ‘lower slope’ and ‘slope base’ in sequential order. Palsa slopes as a whole varied in length with 5 – 8 plots, except for one, which consisted of 3 plots. When possible, the first two plots were assigned to the slope section ‘palsa top’ and the last two plots to the slope section ‘slope base’. The remaining plots were distributed as evenly as possible between the slope sections ‘upper slope’ (closer to the ‘palsa top’) and ‘lower slope’ (closer to the ‘slope base’). This way slope sections had similar sizes and homogenous scattering could be obtained. Palsa

slopes as a whole are referred to as ‘palsa slopes’, whereas the sections along palsa slopes (i.e. palsa top, upper slope, lower slope and palsa base) are referred to as ‘slope sections’.
If active layer was not measured within 1.6 m, active layer was defined as ‘fully thawed’ and described as ‘2 m’ to set a clear difference to all other active layer depths but be able to keep data as numeric. The number chosen for ‘fully thawed’ may however affect the relative importance of active layer depth in the model used later in the analysis.
If sedge species could not be determined in the field, they were later identified with a northern European vegetation handbook (Mossberg 2008) and a European key (Tutin 1980). Several sedge species only occurred in one plot, and it was difficult to determine how much percent was which species if several sedges appeared in one plot. Therefore, sedges were summarized as one group for the main analysis. *Trientalis europaea, Arctostaphylos alpina, Vaccinium oxycoccos* and *Betula pubescens* appeared only once or very few times, so that they could not contribute to patterns in species composition but decreased precision of the analysis and were excluded from species composition analysis. Soil moisture was calibrated using peat from palsas, that were taken out and airdried progressively by Arwen Henz to calculate it as volumetric water content.

2.4.2 Thermokarst

Biomass was washed and separated in living *Sphagnum,* sedges and sedge roots. Living biomass was defined as ‘green’ *Sphagnum* and ‘white’ sedge roots. Thick, straight, root-like structures (~0.5 cm) were excluded, since they may have been former aboveground stem around which the peat grew. Aboveground biomass was dried during the field campaign in an oven (Beketal 701, BDA-15) and later dried again right before measuring weight. Roots were frozen during the field campaign to scan them later using a flatbed scanner (EPSON Perfection V700; SEIKO EPSON CORP, Japan) and the analysing software WinRhizoTM Reg 2008 (Regent Instruments Inc., Canada). They were positioned so that they wouldn't overlap in the tray. Root length, diameter and surface area were measured and afterwards roots were dried and weighed. With the dry weight, the root-specific length and root specific-surface area were calculated.
Sphagnum species were identified with a microscope, a German and a global key (Dierßen 1996; Michaelis 2019). Sedges have their highest rate of growth during June and July (Sloan et al. 2016) and effects from sedges on methane fluxes may change over the growing season. Therefore, data from the period between July 22, and September 2, 2025, were selected to compare methane fluxes from patches with and without sedges.

2.5 Data analysis

The entire data analysis was performed using R Studio Version 2024.04.2+764 and R version 4.4.1 (2024-06-14) (R Core Team, 2024). If required, data were transformed (logarithm, square root and Box-cox transformation) to meet the assumption of normal distribution and homoscedasticity. Box-cox transformation was used on methane fluxes (MASS package; Venables and Ripley 2002). This was the best transformation, that was found, however, assumption of normal distribution was not entirely met indicated by p-values of Shapiro test on normal distribution (sedge $p = 0.09$; non-sedge $p = 0.004$).

Hypothesis 1 on the effect of vegetation height, i.e. *B. nana* versus low vegetation, on active layer depth was tested using the difference of paired values in a one-sample t-test. Kruskal-Wallis test and post-hoc Dunn test were conducted to test differences between the vegetation cover of all occurring vascular plants in *B. nana* and low vegetation patches. Active layer depth and vegetation height was pearson correlated for *B. nana*.

To test **Hypothesis 2** on differences in species composition among plots, the Bray-Curtis three-dimensional distance matrix was calculated with metaMDS-function from the vegan package (Jari Oksanen et al. 2025; Version 2.7-2). Scattering of the distance matrix, tested with betadisper-function and permutest, was similar between the slope sections (palsa top, upper slope, lower slope and slope base). To analyse differences in species composition along the palsa slopes, the distance matrix was used in a permutation analysis of variance (permanova) calculated with the andonis2 function (vegan package). Slope sections and active layer depth were added as explanatory variables in the permanova model. To compare the differences in species composition between slope sections, a pairwise permanova was performed (pairwise.perm.manova function; RVAideMemoire package; Maxime HERVE 2025; Version 0.9-83-12). Similarity percentages (simper function; vegan package) were analysed to evaluate how much single species contribute to the differences in species composition that were analysed beforehand (Clarke 1993). For visualization of species relative abundance along the slope sections, bar-plots were created using ggplot2 (Wickham 2016).

Alpha-diversity was analysed with the Shannon-Wiener Index H and evenness E (vegan package). Additionally, the mean number of species and total number of species (species richness) in the slope sections were calculated. Beta-diversity was calculated with the package betapart (Baselga et al. 2025) for the slope sections, for the slopes within the study sites and all plots. The total beta-diversity (Sørensen-dissimilarity index) consists of the turnover (Simpson-dissimilarity Index) and the nestedness component. The highest total beta-diversity is represented by a value of 1 at completely different species compositions across sites. The

turnover component depicts replacement of species by one another. The nestedness component describes to which degree species assemblage is a subgroup of a species composition at another location.

For analysing **Hypothesis 3** species composition, relative abundance of species in every vegetation zone was calculated by weight and visualized with ggplot2 (Wickham 2016). Differences in biomass and root morphology were analysed for the three sampling locations using one-way ANOVA and post hoc test Tukey. *Sphagnum* weight was tested with a student's t-test, since there were only two locations with *Sphagnum spp.* in the biomass analysis. Methane fluxes of sedge and non-sedge patches were compared with a two-sample t-test.

3. Results

3.1 Palsa

Consistent with the **first hypothesis**, the active layer beneath *B. nana* was nearly 10 cm shallower than beneath low vegetation ($p < 0.001$; Figure 5). Mean active layer depth under *B. nana* was 43.9 cm and under low vegetation 53.4 cm. However, there was no relationship between vegetation height of *B. nana*, ranging from 8.0 to 41.3 cm, and active layer depth (37.0 to 53.5 cm) (Appendix Table A 1; Figure A 1: Active layer depth (Y-axis, cm) against vegetation height (Y-axis, cm) with low vegetation in orange and *B. nana* in turquoise. Correlation and regression were done for *B. nana* only (pearson correlation).Figure A 1). Vegetation height was 18.5 cm higher in *B. nana* plots than in low vegetation plots ($p < 0.001$). Both *B. nana* and low vegetation patches included *Empetrum nigrum, Vaccinium uliginosum, Vaccinium vitis-idea, Andromeda polifolia, Rubus chamaemorus,* bryophytes and lichen. Only *B. nana* and *Rhododendron tomentosum* cover differed between *B. nana* and low vegetation patches (Appendix Table A 2).

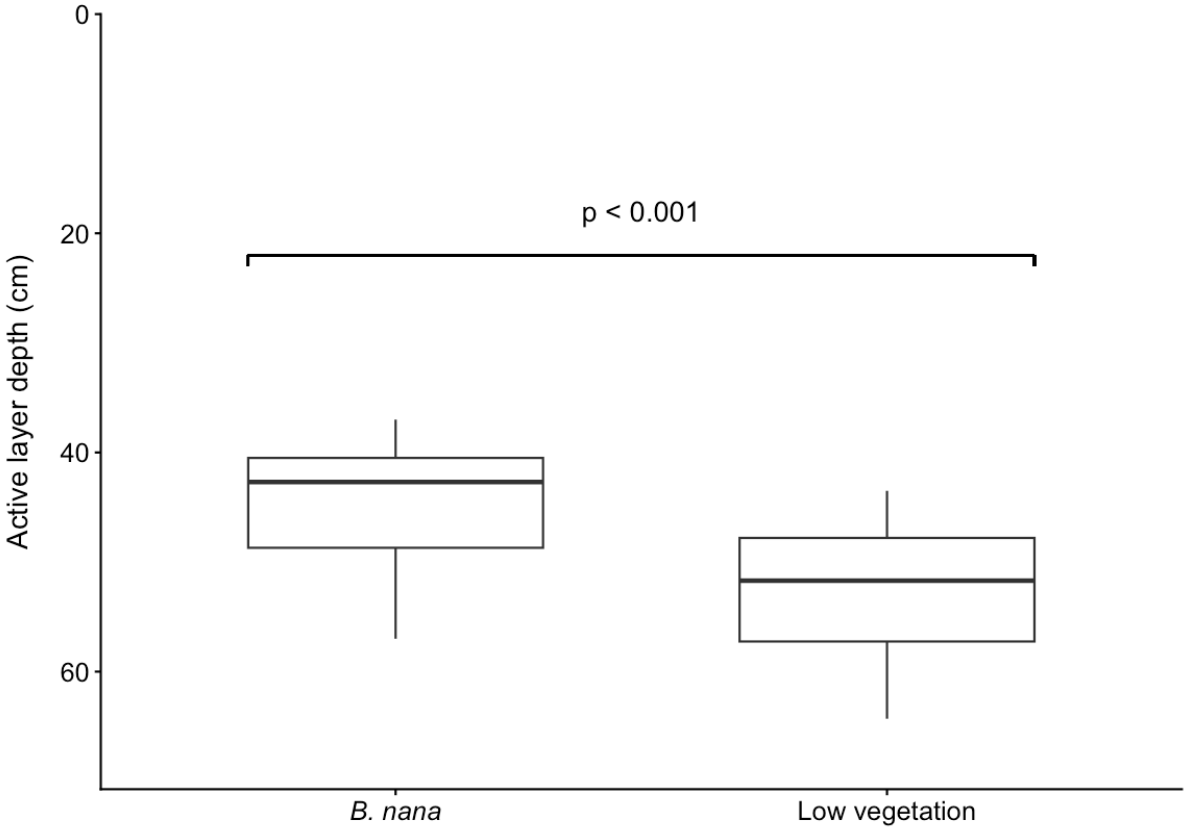


Figure 5: Box plot of active layer depths under *Betula nana* and low vegetation on palsa plateaus in Ádejávri between late August and the beginning of September 2025 (n = 19 paired *B. nana* and low vegetation). Active layer depth under *B. nana* was 43.9 ± 5.07 cm (mean ± sd) and under low vegetation 53.4 ± 6.21 cm (mean ± sd). Data was logarithm transformed for the t-test to meet the assumption of normal distribution.

3.2 Plant gradient along palsa slopes

The **Hypothesis 2.1**, that species composition changes along the slope sections, can only partially be confirmed. Species composition only differed significantly between the slope base and the other slope sections, while the remaining sections did not differ significantly from one another (Appendix Table A 4). The model with slope sections and active layer depth as explanatory variables explained 19% of the difference in species composition (p = 0.001; Table 2). Slope sections explained 12% (p = 0.003) and active layer depth 4% (p = 0.025) of the difference in species composition (Table 2). Species composition generally varied between palsa slopes as a whole (p = 0.001) (Table A 3 Appendix).

Table 2: Results from permanova on differences in plant species composition among the slope sections. R^2 indicates the extent to which the model can explain the overall differences in species composition (model) and the extent to which these differences are explained by the explanatory variables 'slope sections' and 'active layer depth'.

Model variables	Degrees of freedom	Sum of squares	R^2	F	P
Slope section	3	0.52	0.12	2.69	**0.003**
Active layer depth	1	0.17	0.04	2.58	**0.025**
Model	4	0.86	0.19	3.31	**0.001**
Residual	56	3.63	0.81	-	-
Total	60	4.49	1	-	-

A few species contributed significantly to the differences in species composition across slope sections. For the comparison of palsa top to slope base, total plant cover contributed 5% to differences in species composition and plant species *Empetrum nigrum* contributed 10%, *Rubus chamaemorus* 7% and sedges 2%. Sedges show a high standard deviation (sd = 0.04), which is due to the fact that the sedges exclusively but not always occur at the slope base. For the comparison of upper and lower slope to slope base *Rubus chamaemorus* significantly contributed 7% to the differences respectively. Bryophytes contributed 9% to the differences between upper slope and slope base (Table 3). The relative abundance of *Empetrum nigrum* gradually decreased while *Rubus chamaemorus* increased from palsa top to slope base (Figure 6). *Vaccinium myrtillus* was absent at the top but was frequent in base plots and rare along the slope (Figure 6). Sedges only grew at the slope base, in the transition to thermokarst, however with different species of sedges at different palsa slopes. Sedge species *Eriophorum vaginatum, Eriophorum russeolum, Carex rotundata* and *Carex stenolepis* were found several times while *Eriophorum angustifolium, Carex rariflora* and *Carex glacialis* occurred once (Appendix Table A 6). *Betula pubescence* (1 occurrence)*, Vaccinium oxycoccos* and *Trientalis europaea* (1 occurrence) were found at the slope base, while *Arctostaphylos alpina* occurred once at the palsa top.

Table 3: Species that contributed significantly to differences in species composition between slope section comparisons (only includes comparisons of slope sections that differed in former analysis), their average contribution to differences in species composition ('contribution' mean ± sd) and estimated probability of getting similar or higher contribution in random permutation. These were results of the similarity percentage analysis (simper; vegan package in R).

Species/ Total cover	Comparison	Contribution (mean ± sd)	P
Empetrum nigrum	Palsa top – Slope base	0.10 ± 0.06	**0.025**
Rubus chamaemorus	Palsa top – Slope base	0.07 ± 0.06	**0.001**
Sedges	Palsa top – Slope base	0.02 ± 0.04	**0.001**
Bryophytes	Palsa top – Slope base	0.09 ± 0.07	0.059
Total cover	Palsa top – Slope base	0.05 ± 0.05	**0.009**
Rubus chamaemorus	Upper slope – Slope base	0.07 ± 0.06	**0.001**
Bryophytes	Upper slope – Slope base	0.09 ± 0.07	**0.001**
Rubus chamaemorus	Lower slope – Slope base	0.07 ± 0.06	**0.004**
Bryophytes	Lower slope – Slope base	0.10 ± 0.08	0.057

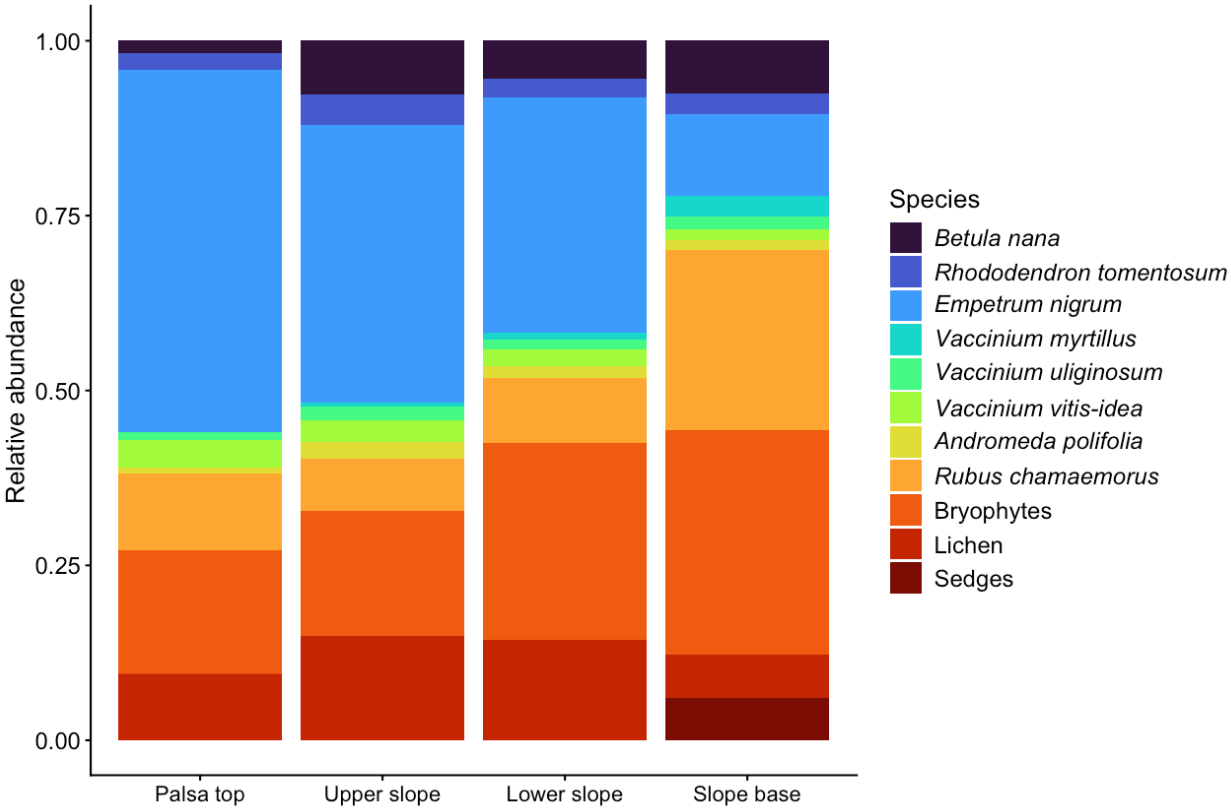


Figure 6: Relative abundance of plant species in the sections of the palsa slope from palsa top to slope base. Very rare species were not visible in the bar plot and are thus excluded. For more details see Table A 6 Appendix.

Hypothesis 2.2, that plant species diversity is highest at the base of palsas, and lowest at the palsa top, could only partially be confirmed. There was no difference in Shannon-Wiener Index and evenness, but a higher beta-diversity (Sørensen-dissimilarity index) at the slope base compared to the palsa top (Table 4; Table 5). These patterns are also reflected in the species richness which was highest at the slope base with 20 species. All other slope sections had a total number of ten species. When comparing average number of species in the slope sections, there was no difference. High beta-diversity at the slope base was mainly attributed to a high turnover of species and only to a little degree to nestedness. The palsa top showed lower beta-diversity than the slope base, however compared to all other slope sections, beta-diversity was second highest at the palsa top. At the palsa top, nestedness was most important, the turnover of species (Simpson-dissimilarity) was accordingly low. Total beta-diversity for all plots was 92% with most of the dissimilarity explained by substitution of species (87%) rather than nestedness (5%) (Table 5). Beta-diversity as a whole barely differed between study sites, however, nestedness was higher and turnover lower in Ádejávri compared to Iškoras and Karlebotn. At Iškoras and Karlebotn site substitution was more relevant (73% and 69%).

Table 4: Plant diversity indicators Shannon-Wiener Index, evenness, sum and mean number of species for each slope section.

Species number and diversity indices	**Palsa top**	**Upper slope**	**Lower slope**	**Slope base**
Shannon-Wiener Index (mean ± sd)	1.12 ± 0.344	1.21 ± 0.357	1.18 ± 0.340	1.31 ± 0.331
Evenness (mean ± sd)	0.651 ± 0.133	0.640 ± 0.158	0.643 ± 0.129	0.696 ± 0.135
Total number of species	10	10	10	20
Mean number of species	5.7	6.6	6.3	6.6

Table 5: Beta-diversity for slope sections, study sites (Ádejávri, Iškoras and Karlebotn) and all plots. Sørensen-dissimilarity is the total beta-diversity, Simpson-dissimilarity is the turnover component describing the amount of substitution by other species and nestedness the degree to which species appear in subgroups of other species compositions.

Slope section	Sørensen-dissimilarity	Simpson-dissimilarity	Nestedness component
Top	0.711	0.478	0.233
Upper Slope	0.68	0.552	0.127
Lower Slope	0.591	0.424	0.167
Base	0.834	0.773	0.061
Site			
Ádejávri	0.779	0.62	0.158
Iškoras	0.787	0.726	0.06
Karlebotn	0.794	0.699	0.94
All Plots	0.918	0.868	0.049

3.3 Vegetation composition at a thermokarst in Ádejávri

In alignment with the first part of **Hypothesis 3.1** the plant species composition varied in the thermokarst, with one or a few species often dominating in individual patches (Figure 7). Vegetation consisted mainly of *Sphagnum spp.* but also of non-*Sphagnum* mosses, a liverwort, a sedge and a vascular dwarf shrub (Table 6). The sedge *Eriophorum scheuchzeri* was found in almost all thermokarst patches but with different abundance. Non-*Sphagnum* mosses (*Polytrichum strictum, Pohlia sphagnicola, Warnstorfia fluitans, Hamatocaulis vernicosus*), liverwort (*Odontoschisma sphagni*) and dwarf shrub (*Vaccinium oxycoccos*) grew only in up to three vegetation zones. Species compositions from the seven sampling locations can be organized in five groups after dominating species. Group 1 (AT1) was dominated by *Eriophorum scheuchzeri* (100%), the only group without any *Sphagnum*. In group 2 (AT2, AT3) *Sphagnum riparium* dominated with 98% (AT2) and 89% (AT3), followed by *Eriophorum scheuchzeri* and another *Sphagnum* species each (*S. cuspidatum* (AT2) and *S. lindbergii* (AT3)). Group 3 (AT5) was dominated by *S. lindbergii* (81%) followed by *S. cuspidatum* (18%). In group 4 (AT4, AT6) *S. russowii* (>50%) dominated and in group 5 (AT7) *S. pulchrum* (48%). In group 4 and 5 most species occurred with eleven and nine species respectively and were the only groups in which dwarf shrub *V. oxycoccos* occurred.

The second part of **Hypothesis 3.1**, that biomass and root morphology varies within vegetation zones of the thermokarst, could partially be confirmed. Total and aboveground plant biomass was smallest at vegetation zone AT1 ($p < 0.001$). Sedge (*E. scheuchzeri*) total and aboveground

biomass was highest in AT3 ($p < 0.001$). Sedge root biomass differed between AT3 and AT1 but not between AT3 and AT5. *Sphagnum* biomass almost increased from AT3 to AT5 ($p = 0.05$) (see Appendix Table A 7). Root morphology did almost not differ between vegetation zones. Only diameter increased from 0.53 mm in AT1 to in 0.7 mm in AT3 ($p = 0.003$). As a result of higher root biomass, surface area also increased ($p=0.02$) from AT1 to AT3, but specific root area stayed the same.

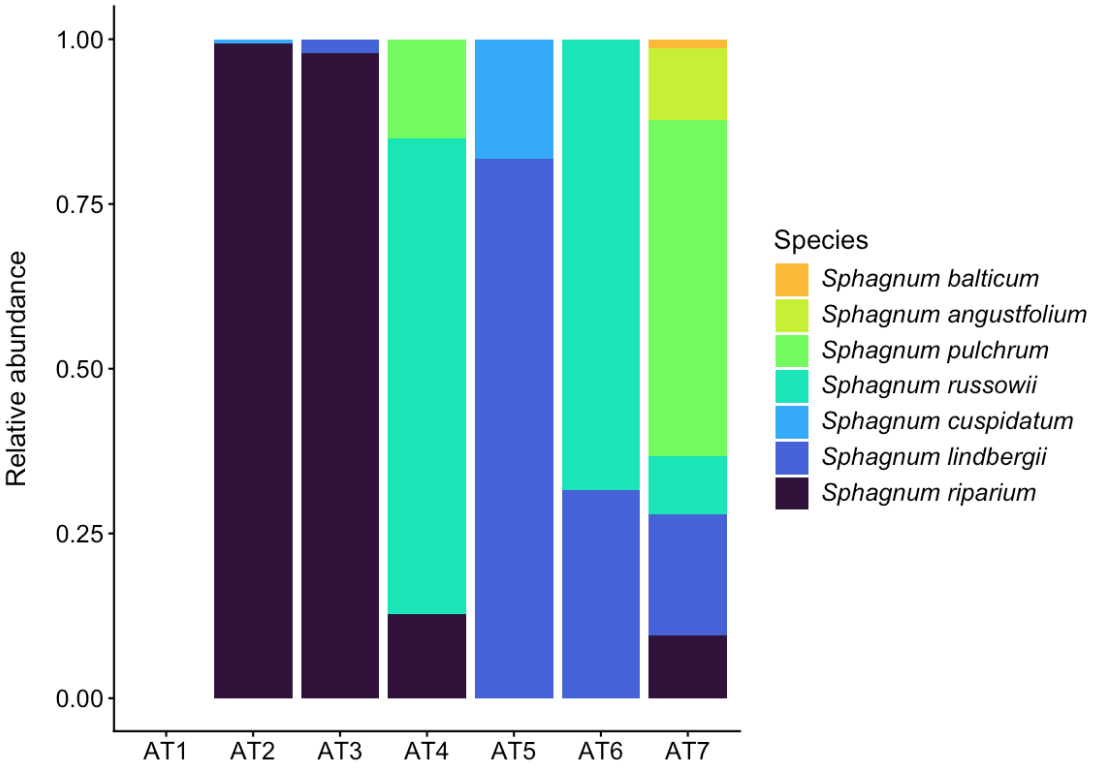


Figure 7: Relative abundance of *Sphagnum spp*. in vegetation zones AT1 to AT7 of the thermokarst in Ádejávri.

Table 6: Relative abundance (%) of all species occurring in vegetation zones AT1 to AT7 at Ádejávri thermokarst.

Species	**AT1**	**AT2**	**AT3**	**AT4**	**AT5**	**AT6**	**AT7**
Sphagnum riparium	0	97.5	88.8	9.9	0	0	9.1
Sphagnum lindbergii	0	0	1.7	0	80.7	27.7	17.4
Sphagnum cuspidatum	0	0.6	0	0	17.9	0	0
Sphagnum russowii	0	0	0	55.9	0	60.2	8.4
Sphagnum pulchrum	0	0	0	11.6	0	0	48.2
Sphagnum angustifolium	0	0	0	0	0	0	10.4
Sphagnum balticum	0	0	0	0	0	0	1.3
Polytrichum strictum	0	0	0	1.9	0	0	0
Pohlia sphagnicola	0	0	0	16.0	0	0	0
Warnstorfia fluitans	0	0	0	0	0	0	0.1
Hamatocaulis vernicosus	0	0	0	0	0	9.6	0
*Odontoschisma sphagni**	0	0	0	0.3	0	0	0
*Odontoschisma sphagni***	0	0	0	0	0	0	4.5
Eriophorum scheuchzeri	100	1.9	9.3	3.7	1.4	0.3	0
Vaccinium oxycoccos	0	0	0	0.8	0	2.3	0.6
Total number of species	1	3	3	8	3	5	9

*with undetermined *Sphagnum* **with *Sphagnum lindbergii*

In alignment with **Hypothesis 3.2** methane emissions in sedge patches were 31 mmol m^{-2} d^{-1} and in non-sedge patches 9 mmol m^{-2} d^{-1} measured between July 22, and September 2, 2025. Mean emissions were 3.4 times higher in sedge patches compared to where sedges have been removed ($p < 0.001$). Emissions strongly varied among chambers and in time. Maximum methane emissions reached up to 336 mmol m^{-2} d^{-1} in sedge patches and up to 88.2 mmol m^{-2} d^{-1} in non-sedge patches.

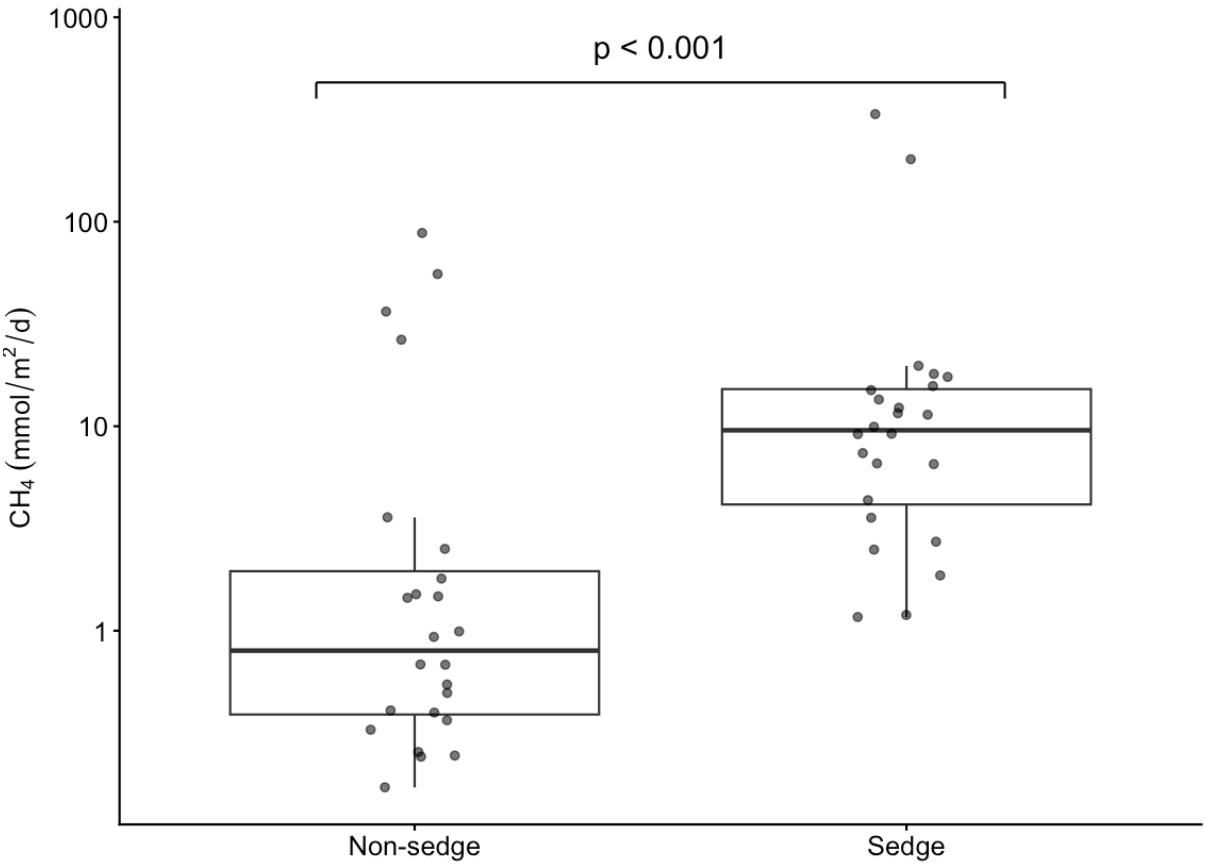


Figure 8: Methane emissions from non-sedge and sedge patches in a thermokarst in Ádejávri from July 22 to September 2, 2025. Boxplots show median (centre line), 0.25 and 0.75 percentile (lower and upper margin of boxes) of methane emissions. Mean emissions were 9 mmol m^{-2} d^{-1} in non-sedge and 31 mmol m^{-2} d^{-1} in sedge patches. Note that the Y-axis is logarithmic. For the t-test data was Box-cox transformed to better meet the assumption of normal distribution.

Differences in methane emissions between sedge and non-sedge patches were low at the beginning of July (start of measurements) and highest in August, where emissions in sedge patches peaked. Soil temperature increased from beginning to mid-July and decreased in the beginning of August. Soil temperature was lowest towards the end of the growing season in late August and beginning of September (Figure 9).

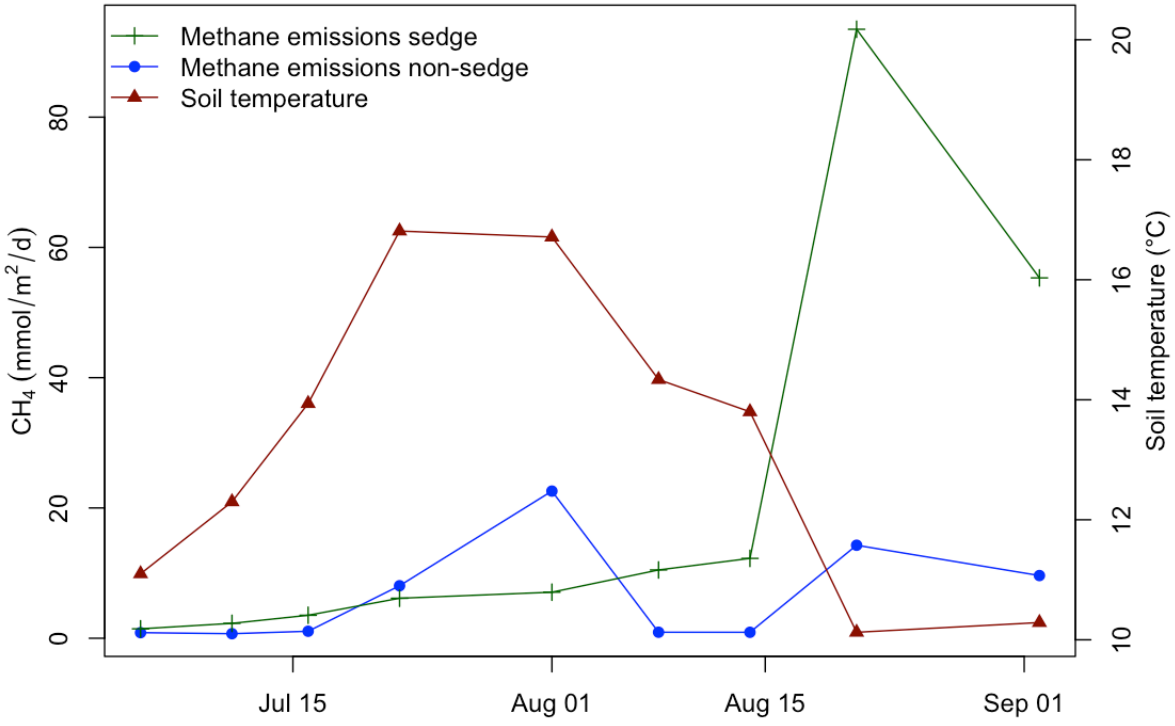


Figure 9: Mean methane emissions (mmol m^{-2} d^{-1}) for sedge (green) and non-sedge (blue) patches and average soil temperature (°C) of sedge and non-sedge patches (red) for each measuring date during the field campaign (2025).

4. Discussion

4.1 Effect of *B. nana* on active layer depth

In accordance with hypothesis 1 mean active layer on palsa plateaus was 10 cm shallower under *B. nana* compared to low vegetation towards the end of the growing season. This aligns with most other studies that found reduced active layer under *B. nana* and other taller shrubs or forests (Blok et al. 2010; Cameron et al. 2024; Yi et al. 2007). Active layer was nine percent deeper in plots with experimentally removed *B. nana* (Blok et al. 2010). Removal of taller shrubs (>40 cm) also resulted in deeper active layer compared to control plots (Cameron et al. 2024). A modelling study showed reduced thaw under both shrubs and forest, where the reduction of thaw was stronger under forest (Yi et al. 2007). Most studies align with the results, showing shallower active layer beneath shrubs.

4.1.1 Shrub-effects in winter

Permafrost is affected by year-round average conditions affecting permafrost temperature (Serreze et al. 2000), which can be complex and contrasting over the course of the year. In winter, deep snow can isolate soil and was found to be an important factor contributing to permafrost degradation, as air temperature alone could not explain warming of permafrost (Serreze et al. 2000). Shrubs can trap snow with protruding branches reducing wind speed (Essery and Pomeroy 2004). Snow in shrubs is more isolative compared to non-shrub affected snow, since it is deeper, less dense and has a higher percentage of depth hoar (Domine et al. 2016; Sturm et al. 2001). This increases soil temperature during winter underneath shrubs compared to non-shrub patches (Aguirre et al. 2021) in some cases by several degrees (Sturm et al. 2001). While experimentally increased snow caused accelerated active layer deepening (Johansson et al. 2013), snow trapping in shrubs has not caused deeper active layer (Cameron et al. 2024). Underneath shrubs counteracting summer effects were often more important than increased winter soil temperatures (Cameron et al. 2024; Yi et al. 2007; Fisher et al. 2016; Aguirre et al. 2021). In one study reduced albedo from protruding shrub branches compared to pure snow led to faster snowmelt in spring, which resulted in deeper active layer even at the end of the growing season (Wilcox et al. 2019). This may vary with snow depth, since deeper snow and faster snowmelt beneath shrubs can cancel each other out and result in simultaneous end of snowmelt in shrub and non-shrub patches (Wilcox et al. 2019; Sturm et al. 2005). Additionally, earlier snowmelt and deeper active layer beneath shrubs at the beginning of the

growing season may not always lead to deeper active layer beneath shrubs at the end of the growing season depending on magnitude of counteracting summer effects (see below 4.1.2 Shrub-effects in summer). While shrubs affect soil temperature via snow trapping, and shrub albedo affects snow melt, this did not deepen active layer underneath shrubs in most studies (Sturm et al. 2001; Aguirre et al. 2021; Wilcox et al. 2019; Sturm et al. 2005; Cameron et al. 2024; Fisher et al. 2016).

4.1.2 Shrub-effects in summer

In summer shrubs can absorb much of the radiation thereby reducing the warming in the top peat during the growing season (Cameron et al. 2024; Yi et al. 2007; Blok et al. 2010; Fisher et al. 2016). In general, high summer temperatures and precipitation are important drivers of permafrost degradation associated with climate change (Jorgenson et al. 2006). Thus, shrubs mitigating temperature and precipitation by reducing soil temperature and moisture are relevant effects for active layer depth (Cameron et al. 2024; Fisher et al. 2016). Active layer depth increased primarily in shrub removal on warm days (°C not specified) and caused differences between *B. nana* and *B. nana* removal (Blok et al. 2010). At Šihččajávri meteorological station close to Ádejávri maximum air temperatures were above 20°C reaching up to 28°C for at least 16 days in July and four in August using data from the Norwegian Meteorological Institute. This might have caused or intensified active layer deepening in low vegetation compared to *B. nana*. The soil heat flux was lower for *B. nana* compared *B. nana* removal especially during warm periods as well as ratio of soil heat flux and net radiation (Blok et al. 2010). This was argued to be a result either of more shading or higher evapotranspiration and higher percentage of latent heat flux, which was also found for taller shrubs and forest canopy (Blok et al. 2010; Fisher et al. 2016; Cameron et al. 2024). In addition shrubs also reduced mid-summer soil moisture (Aguirre et al. 2021). The reduction in water content was attributed to water uptake by roots and transpiration increasing with leaf area index. This reduces thermal conductivity in soils (Fisher et al. 2016). Low water content in organic soils in summer can insulate the permafrost table from warm air temperatures well, due to their high pore volume thus high percentage of isolating air (Yi et al. 2007). Here shrubs likely protected permafrost through shadowing and higher transpiration reducing the downward heat flux and soil moisture of the organic soil.

4.1.3 Other factors influencing active layer depth

Active layer depth at Ádejávri did not correlate with height of *B. nana* (8-41 cm) (Appendix Figure A 1), which is contrary to what other studies have found (Blok et al. 2010; Fisher et al. 2016). Higher leaf area index increases transpiration and shadowing and was associated with shallower active layer (Fisher et al. 2016). Across Ádejávri study site variation in active layer depth was high, which might be due to other important factors besides height of shrubs affecting active layer depth. Factors such as soil moisture, peat and moss layer thickness are important for reduced active layer deepening (Fisher et al. 2016; Smith et al. 2009). Organic layers in permafrost soils reduce thaw, due to the high pore volume, which can store more ice and can insulate permafrost well under dry conditions (Yi et al. 2007; Fisher et al. 2016). High soil moisture however can offset insulation due to higher heat conductance (Fisher et al. 2016). Similarly dry moss layers insulate soil well (Turetsky et al. 2012; Fisher et al. 2016) and moss can contribute to higher latent heat flux with high evaporation (Beringer et al. 2005; Blok et al. 2010), both contributing to lower soil heat flux with thicker moss layers. General variability in the degradation status of palsas or varying distance to palsa slope could also affect differences in active layer depth, however, to avoid this, it was taken care to keep some distance from palsa slopes. Soil moisture, organic layer and moss layer thickness were not measured in this study but based on what others found they could be important factors besides shrub height for variation in active layer depth (Fisher et al. 2016; Smith et al. 2009). Across the study site variation in active layer depth from soil moisture, organic layer and moss layer thickness may have been more pronounced than shrub height and may explain why there was no correlation between *B. nana* height and active layer depth.

4.1.4 Shrub-effects on active layer depth in a changing climate

Shrubs have been increasing with warming in the arctic (Mekonnen et al. 2021; Myers-Smith et al. 2011; Elmendorf et al. 2012) and snow-free season is expected to extend (Yi et al. 2007). This may increase differnces in active layer depth between shrubs and low vegetation, since summer effects and especially warm periods have been found to be important for shrubs hampering permafrost thaw (Cameron et al. 2024; Blok et al. 2010). Additionally, formation of ice layers during short warm spells in winter especially beneath shrubs due to lower albedo may increase with ongoing climate change. This could increase winter ground cooling beneath shrubs and reduce isolation of snow in shrubs (Barrere et al. 2018; Hansen et al. 2014). For a detailed understanding of processes leading to the shallower active layer under *B. nana* in this

study further measurements are necessary. Based on what others have found the 10 cm shallower active layer under *B. nana* probably results from shadowing, water uptake and higher transpiration rates and thus reduced soil temperature and moisture under *B. nana* in summer. These effects may become more relevant in further climate warming, since shrubs are expected to further expand and temperatures to rise, until exceeded by thaw.

4.2 Plant gradient along palsa slopes

In line with hypothesis 2.1, species composition, including plant species abundances, differed at the slope base compared to the other slope sections. Mostly sedges, *Empetrum nigrum*, *Rubus chamaemorus* and total vegetation cover contributed to those differences in the species composition. Slope section and active layer depth together explained 19% of the variability in the species composition. Sections (12%) were more important than active layer depth (4%) in explaining species composition, indicating that other factors associated with slope section affect plant species composition (see below 4.2.1).

4.2.1 Slope sections

In the sections from palsa top to palsa base, permafrost degradation and active layer depth increased. Active layer deepening is influenced by both higher temperature and soil moisture conducting more heat to the permafrost table (Jorgenson et al. 2006). Inversely, active layer deepening causes ground subsidence and water can accumulate (Schuur et al. 2007; Schuur et al. 2015). In contrast, palsas plateaus are mostly dry and soil moisture is precipitation dependent due to runoff and drainage at the steep slopes (Zuidhoff and Kolstrup 2005). In addition, the peat surface may become hydrophobic during summer phases without rain, which could increase the surface runoff from the palsa plateaus, moreover high radiation and strong winds could increase evapotranspiration. During winter, strong winds can reduce snow cover, resulting in colder temperatures at the palsa surface and reduced water infiltration during snowmelt (Zuidhoff and Kolstrup 2005; Martin et al. 2021). Soil moisture and soil temperature measurements did not support the findings literature, however since measurements were only taken once, explanatory power is limited. Measurements only indicate lower soil temperatures at the slope base compared to the upper slope (Appendix Figure A 2; Figure A 3). Taken together, soil temperature, volumetric water content and active layer deepening are related and likely change in the sections from palsa top to slope base and consequently may affect species composition.

4.2.2 Active layer depth

Active layer deepening was suggested to favour deep rooting sedges and *Rubus chamaemorus* by increasing root-penetrable soil volume and plant-available nitrogen in form of ammonium at the thaw front (Blume-Werry et al. 2019; Keuper et al. 2017). This may at least partially explain second highest abundance of *Rubus chamaemorus* at the top of palsa. *R. chamaemorus* was reported to have a maximum root depth of 67 cm in non-permafrost tundra soils (Jasinski et al. 2022). This root depth exceeds that of most other plant species, as more than 90% of the root biomass in the tundra biome is found in the upper 30 cm soil depth (Jackson et al. 1996). For example, roots of *Empetrum nigrum* grow mainly in the upper 10 - 15 cm soil depth (Iversen et al. 2015; Michelsen et al. 1996). Thus, *R. chamaemorus* is adapted by growing roots in soil volume with little to no resource competition (Taylor 1989), which is likely advantageous here. It was, additionally shown that *R. chamaemorus* took up experimentally increased inorganic nitrogen close to the thaw front at 45 and 50 cm depth and increased root biomass by up to 100% (Keuper et al. 2017; Blume-Werry et al. 2019). At Ádejávri ammonium concentrations were considerably higher at soil depths of 40 - 60 cm (1.1 mmol/l KCl extracts) compared to 0 – 40 cm soil depth (0.6 mmol/l KCl extracts) in palsas (A. Gnaedinger (personal communication)). Mean active layer depths from all three study sites were 64 cm, 100 cm and 161 cm for palsa top, upper slope and lower slope respectively. Thaw depths at the palsa top and depth of increased ammonium concentrations are within the range of reported rooting depths of *R. chamaemorus*. This may at least partially explain, why *R. chamaemorus* had its second highest abundance at the palsa top and was the second most abundant vascular plant at the top. Additionally, *R. chamaemorus* is tolerant to strong winds and thin to absent snow cover common at the top of palsas (Whittaker 1993; Zuidhoff and Kolstrup 2005). Thus, second highest abundance of *R. chamaemorus* at the palsa top may at least partially relate to increased nutrients close to the thaw front.

At the slope base, *R. chamaemorus* had its highest abundance and was the most common vascular plant species. Conditions were likely wet or water-saturated due to its close proximity to the thermokarst, which usually accumulated water, also indicated by the high abundance of *Sphagnum spp.* and of hydrophilic sedges (Tyler et al. 2021). The slope base was the only slope section where sedges grew. Sedge roots can reach soil depths of 1 m and were also shown to benefit from experimentally increased ammonium close to the thaw front (Blume-Werry et al. 2019). At base plots however, permafrost was almost always fully thawed or at least deeper than 85 cm. Thus, the release of nutrients associated with permafrost thaw may be low at the slope base, but moist conditions possibly affect nutrient concentrations and uptake by plants.

For example, at thermokarsts extractable ammonium was found to be higher compared to palsas (Morison et al. 2018). Jasinski et al. (2022) found, that water table was more important than active layer deepening for increased plant canopy nutrients, since for example, *R. chamaemorus* had increased canopy nitrogen and phosphorous concentrations at high water tables. In addition to the effect of potentially increased nutrients, *R. chamaemorus* and sedges are adapted to moist conditions. Aerenchymatous cells in sedges allow oxygen transport from the leaf via stem to the roots, enabling growth in wet habitats (Iversen et al. 2015; Vartapetian and Jackson 1997). Similarly, *R. chamaemorus* has porous structures in roots probably functioning as aerenchyma (Taylor 1971). Thus, high abundances of sedges and *R. chamaemorus* at the slope base may be more related to moisture and moisture effects on nutrient availability rather than to active layer deepening. Overall, increased *R. chamaemorus* abundance at the palsa top likely relates to increased nutrients through active layer deepening, while at the slope base increased moisture and related nutrient availability may be more important for high abundance of *R. chamaemorus* and sedges.

4.2.3 Abundance of *Empetrum nigrum* along palsa slopes

Empetrum nigrum decreased in abundance from palsa top to slope base and dominated with 52% relative abundance at the palsa top. Consistent with that *E. nigrum* was identified as characteristic for stable palsa plateaus in literature (Zuidhoff and Kolstrup 2005). On exposed palsas conditions are harsh with dry peat, high radiation and strong winds, resulting in thin snow cover and lower temperatures at the palsa surface in winter (Zuidhoff and Kolstrup 2005). Resulting stress favours *E. nigrum* in competition, since it is relatively tolerant to the harsh conditions on palsas compared to other plant species. It grows in dry conditions and where the snow cover is thin, as it is tolerant to low temperatures, also supported by its growth in low mats (Bell and Tallis 1973; Whittaker 1993; Zuidhoff and Kolstrup 2005). It is an early colonizer in cold climates and also grows on nutrient poor soils facilitated by mycorrhiza (Whittaker 1993; Tybirk et al. 2000). It can hamper germination, growth and survival of other plant species by competition and with allelopathic chemical compounds. Otherwise, this species can have sheltering effects supporting growth of other species depending on its abundance (Aerts 2010; Nilsson 1994). This may reduce abundance of other species, but *R. chamaemorus* seedlings for example, were also found to benefit from shelter of *Empetrum nigrum,* despite of its hampering effects (Aerts 2010). Almost all vascular plants at the palsa top were ericaceous species associated with ericaceous mycorrhiza or had arbuscular mycorrhiza associations

(*B. nana*) except for *R. chamaemorus* (Tyler et al. 2021; Taylor 1989). Mycorrhiza can enhance resource uptake, and different mycorrhizal associations were found to access different resources of nutrients. For example ericaceous plants can access organic nitrogen via mycorrhiza in organic soils (Makarov et al. 2020; Michelsen et al. 1996). This may enhance the growth of species with mycorrhiza associations. Taken together, *Empetrum nigrum* dominates at the palsa top likely due to little competition in the harsh conditions and it may further reduce competition through the release of allelopathic compounds.

At the upper and lower slope on the edges of palsas, permafrost degradation and ground subsidence is accelerated (Martin et al. 2021). This could explain reduction of *Empetrum nigrum* at the upper and lower slopes of palsas, as it has little tolerance against mechanical disturbance (Tybirk et al. 2000). Even though upper and lower slopes are overall dry due to drainage, water content may not be uniform (Zuidhoff and Kolstrup 2005), which may reduce amount of suitable patches for *E. nigrum*. The formation of cracks during palsa degradation may accelerate drainage around cracks, indicated by lichen-dominated patches at upper and lower slope sections. Non-gradual active layer deepening and ground subsidence on the other hand may reduce surface runoff and drainage and increase moisture in small patches. This is also indicated by the sporadic occurrence of *Vaccinium myrtillus* at the upper and lower slope, which requires mesic to moist conditions (Tyler et al. 2021). Additionally, lower wind velocities at the upper and lower slopes reduce evapotranspiration and increase snow accumulation (Essery and Pomeroy 2004). *B. nana* abundance tended to be higher compared to the palsa top, which aligns with deeper snow at the upper and lower slope, since *B. nana* prefers locations with deeper snow and accumulates snow (Essery and Pomeroy 2004; Sturm et al. 2005; Zuidhoff and Kolstrup 2005). The occurrence of *Vaccinium myrtillis* at upper and lower slopes and slope base, can also be attributed to deeper snow, which protects it from cold temperatures during winter (Erik Sjogren et al. 1999; Zuidhoff and Kolstrup 2005). *Empetrum nigrum* however, is disfavoured under deep and long-lasting snow cover, as it is an early flowering species (Bell and Tallis 1973; Olofsson et al. 2011). Additionally, long-lasting snow can promote an infection by the pathogenic fungi *Arwidssonia empetri*. After extended humid periods under snow *A. empetri* can spread and strongly reduce *E. nigrum* (Olofsson et al. 2011). More shelter along upper and lower slopes may thus disfavour *E. nigrum* while improving growing conditions for other plants. Thus, increasing shelter, deepening snow and soil disturbance along the palsa slopes could potentially reduce abundance of *Empetrum nigrum*.

Conditions at the base were likely wetter or water saturated, also indicated by the increase in mosses and bog species *Vaccinium oxycoccos*, *Trientalis europaea, R. chamaemorus* and by

the sudden appearance of sedges (Erik Sjogren et al. 1999; Tyler et al. 2021). Thus, conditions may be too wet for *Empetrum nigrum,* resulting in its low relative abundance of 12% (Whittaker 1993; Bell and Tallis 1973). Additionally, total abundance was highest at the slope base, which likely reflects the more sheltered conditions (Choler et al. 2001) and increases competition compared to the palsa top (Zuidhoff and Kolstrup 2005). Snow accumulating in thermokarst depressions (Martin et al. 2021) additionally disfavours *E. nigrum* (Bell and Tallis 1973). Deep snow and high moisture at the slope base further increase the risk of the pathogen fungi *A. empetri* potentially damaging *Empetrum nigrum* (Olofsson et al. 2011). Thus, the low abundance of *E. nigrum* at the slope base may be a result of moist conditions, stronger competition and increased snow depths.

Overall, *Empetrum nigrum* decreases gradually from the palsa top with 52% relative abundance to the slope base with 12% relative abundance. This may relate to several biotic and abiotic factors, such as soil moisture, temperature, exposure to wind, soil disturbance, competition and snowpack.

Species composition along the palsa slope shifted at the slope base and was mostly driven by sedges appearing only at the base of the palsa, increasing *R. chamaemorus* and decreasing *E. nigrum*. Changes in species composition were probably a result of a combination of factors, which need further investigation. Based on what others found, they may relate to changes in active layer depth, moisture, nutrient availability, temperature, competition and snowpack along the palsa slopes from exposed palsa top to sheltered slope base.

4.2.4 Biodiversity

In this study, alpha-diversity of plants was similar along the slope sections. According to this, both evenness and average species richness were similar among sections. This contradicts hypothesis 2.2, which states that plant diversity is highest at the slope base and lowest at the palsa top. With degradation of palsas, microtopography and plant diversity were found to increase due to the formation of irregular surfaces and cracks (Zuidhoff and Kolstrup 2005). Though microtopography was not quantified here, at least moisture was likely more variable at the slope base compared to other sections, since it was located at the transition of dry palsa to water saturated thermokarst. A high variability in volumetric water content at the slope base was also indicated by measurements done once (Appendix Figure A 3). Supporting this, Beilman (2001) suggested that alpha-diversity was low on palsas due to a lack in moisture gradients compared to thermokarst. Number of ecological niches were commonly found to be related to alpha-diversity (Yang et al. 2015). In some cases, however, other factors such as

regional species pool, niche size or light were more important or complicated niche-diversity relationships (Lundholm 2009). Here, productivity, indicated by varying total plant cover, may be an important additional factor influencing plant diversity. Higher alpha-diversity in vascular plants has been associated with lower productivity related to lower competition in the arctic tundra (Virtanen et al. 2013). According to this, alpha-diversity should be lowest at the slope base, where total cover was highest. Greater variability in moisture, implying higher number of ecological niches, could on the other hand increase alpha-diversity at the slope base. Thus, productivity and the diversity of ecological niches could be counteracting factors, leading to similar levels of alpha-diversity among sections.

When comparing diversity across palsa slopes at the three study sites, the base of palsas showed highest beta-diversity, which is mainly attributed to a high turnover. This may be mainly due to the high turnover of sedges occurring at different pH at the slope base. According to indicator values suggested by Tyler et al. (2021), the sedges found in this study varied in pH preference from strongly acidic to alkaline habitats. Due to the close proximity of slope bases to thermokarsts, this may at least partially reflect the pH of thermokarsts. For example, at Ádejávri and Iškoras *Eriophorum vaginatum* with strongly acidic indicator values (<4.5) (Tyler et al. 2021) occurred. This aligns with strong acidity from the high amounts of organic matter in peatlands, as reported for thermokarsts in Iškoras (Holmes et al. 2022; Knutson et al. 2025). Additionally, at Iškoras moderately acidic to subneutral (pH 4.5-6.5) sedges (*Eriophorum russeolum, Carex rostrata* and *Carex rotundata*) (Tyler et al. 2021) were recorded. This may be attributed to a wetland stream at Iškoras, which was almost neutral in pH (mean pH >6) because of carbonate and silicate mineral weathering (Knutson et al. 2025). At Karlebotn, sedge species with indicator values from subneutral pH (5.5 to 6.5) (*Eriophorum angustifolium*) to alkaline pH (>7.5) (*Carex glacialis*) (Tyler et al. 2021) occurred. The reason for their occurrence remains unknown, since the surrounding geology does not indicate effects on alkalinity as there is mainly granite gneiss and clay slate bedrock (Geological Survey of Norway 2026). Additionally, Karlebotn is oceanic, but the seashore may be too far away (~4 km distance) for elevating pH (Whittle and Gallego-Sala 2016). In summary, no difference in alpha-diversity along the slope sections was found, likely due to counteracting effects of productivity and variability in moisture. Beta-diversity was highest at the slope base, likely due to high turnover, especially of sedges, which may relate to high pH variability among palsa slopes.

4.3 Thermokarst

4.3.1 Vegetation composition in a thermokarst at Ádejávri

Species composition varied within the thermokarst, as evidenced by five vegetation zones (groups), which represent different successional stages. The confirmation of hypothesis 3.1 suggests that the age of thermokarst formation, often associated with differences in water level, have a major influence on plant composition in a relatively small area. Overall, 14 plant species were identified in the thermokarst groups, thereof seven *Sphagnum* species, one sedge and five other bryophytes and one dwarf shrub. The following discussion examines the differences between the five vegetation groups in greater detail. Group 1 was related to a recently thawed, initial thermokarst stage, group 2 was likely young, group 3 intermediate and group 5 an older successional stage.

The first group consisted mainly of bare, well-decomposed peat and the only occurring species was the sedge *Eriophorum scheuchzeri.* Thermokarst formation by palsa collapse disturbs the physical structure of peat often resulting in death of former vegetation and bare ground (Lantz et al. 2009). In the initial phase, sedges emerge in the bare peat, if the water table is high (Gałka et al. 2017). In the event of significant ground subsidence and high water table, initial thermokarst ponds with an open water surface may also form (Magnússon et al. 2020; Olefeldt et al. 2021), but this was not the case at the studied thermokarst. Sedges were reported to establish within 8 years after thermokarst pond formation (Magnússon et al. 2020) while cover by *Sphagnum* and sedges was observed after approximately 30 years (Heffernan et al. 2022).

In group 2, *Sphagnum riparium* and sedge *Eriophorum scheuchzeri* dominated the species composition. *S. riparium* and sedges typically dominate in early successional stages and belong to the first colonizers in thermokarst ponds (Camill 1999b; Heffernan et al. 2022; Gałka et al. 2017). *S. riparium* is an aquatic species and is associated with high water tables in recently thawed permafrost (Camill 1999b, 1999a). In group 2, *Eriophorum scheuchzeri* was most abundant compared to the other groups of the thermokarst. *S. lindbergii* occurred only as a few plants near group 3. *S. lindbergii* is not typical for this early successional stage in thermokarsts, but more generally occurs with *S. riparium* in northern peatlands (Daniels et al. 1985; Horton et al. 1979). Species composition of group 2 is typical for early successional stages in thermokarsts ponds.

S. lindbergii, a hydrophilous species that is characteristic of group 3, proved to be dominant in the transition from fen to bog as well as in oligotrophic to mesotrophic habitats (Daniels et al. 1985; Gałka et al. 2017). *S. lindbergii* is otherwise ecologically similar to *S. riparium* and forms

loose floating mats (Horton et al. 1979). The dominance of *S. lindbergii* in the transition from fen to bog indicates an intermediate transitional stage of the thermokarst succession.

Samples from group 4, dominated by *S. russowii*, were taken from thermokarst edges. Overall, 11 species were found within group 4, most of which appear in a wide range of habitats. For example, *Odontoschisma sphagni*, a liverwort species and dominant *S. russowii* occur in a wide range of moisture and trophic state (Bosanquet 2020a; Horton et al. 1979). *S. russowii* was found in intermediate and nutrient-poor thermokarst fens and bogs; however, it usually only occurred sporadically and slightly tended to less wet habitats (Horton et al. 1979). There was also the moss *Polytrichum strictum* growing and the ericaceous dwarf shrub *Vaccinium oxycoccos*, which mostly grow in less wet habitats (Rumsey 2020; Guêné-Nanchen et al. 2025). *S. riparium*, S. *lindbergii* and *S. pulchrum*, which occurred in some plots assigned to group 4, are on the other hand typical for wet lawns in cold climates (Daniels et al. 1985; Guêné-Nanchen et al. 2025). The occurrence of species preferring different moisture, may indicate a moisture gradient within group 4. There may be a shallower water table towards the centre of the thermokarst and a deeper water table at the edge of the thermokarst, due to less subsided ground. Additionally, dominating *S. russowii* was not associated with thermokarst successional stages before (Horton et al. 1979). Thus, species composition of group 4 may not be related to a successional stage but rather be a result of variability in moisture at the edge of thermokarst.

Group 5, dominated by *S. pulchrum*, comprised nine species with somewhat varying habitats along moisture and chemical gradients. *S. pulchrum* appears as a dominant species in wet lawns (Guêné-Nanchen et al. 2025), just like second most dominant *S. lindbergii* (Gałka et al. 2017). Similarly, *Warnstorfia fluitans*, which here occurred with little abundance, grows in acidic, nutrient-poor pools filled by rainwater (Bosanquet 2020b). However, also more xeric species like *S. angustifolium* and *S. balticum* were present, which were reported to indicate lower water table and transition to hummocks in bogs (Gałka et al. 2017; Camill 1999b, 1999a). *S. balticum* was found in old successional stages where thermokarsts were already decreasing (Magnússon et al. 2020). At those older successional stages, indicated by *S. balticum*, dwarf shrubs were found to establish, due to moist but not saturated conditions (Magnússon et al. 2020). Here, the dwarf shrub *V. oxycoccos* was found and no *E. scheuchzeri*, which was found in all other groups. This fits other findings, since *Carex spp.*, which is functionally similar to *Eriophorum spp.*, was negatively correlated with *S. angustifolium* (Camill 1999a). *S. lindbergii* is not a successful competitor when peat rises further up the water table and quickly disappeared with the occurrence and growth of *S. balticum* (Gałka et al. 2017; Camill 1999b). This explains why here *S. lindbergii* wasn't found in samples, where more xeric species *S. angustifolium* and

S. balticum appeared. When peat rises further above the water table *S. angustifolium* gets replaced by hummock forming *S. fuscum*, associated with late succession in thermokarsts (Horton et al. 1979; Camill 1999b). *S. fuscum* did however not occur here. Thus, group 5 may relate to an older successional stage at the transition to lower water tables, indicated by *S. angustifolium*, but the absence of *S. fuscum* and the occurrence of both hydrophilic and more xeric species.

Overall, there was variation in species composition within the thermokarst in Ádejávri, which may relate to successional stages in the thermokarst, except for group 4. Groups 1 - 5 were identified as initial, young, intermediate and old thermokarst stages respectively. These classifications were only based on observed vegetation and literature and need further dating of time passed since thaw, to be confirmed.

4.3.2 Biomass and root morphology

The second part of hypothesis 3.1, that living biomass and root morphology vary within thermokarst and are related to the successional stages, could only partially be confirmed. Due to the initially low vegetation (Lantz et al. 2009) total biomass increased from group 1 to the other two vegetation zones group 2 and 3, while total biomass of group 2 and 3 was similar. In group 2 sedge biomass was highest, typical for young successional stages in thermokarsts associated with *S. riparium* (Gałka et al. 2017; Heffernan et al. 2022). Also above to belowground ratio of sedge biomass was highest in group 2 (Appendix Table A 7), which may be related to nutrient availability according to the functional biomass allocation hypothesis (Poorter et al. 2012). This hypothesis assumes that plants allocate more carbon belowground and can thus acquire more nutrients at low nutrient availability. In anaerobic sedge peat nitrogen mineralization rates were much higher than in *Sphagnum* peat (Updegraff et al. 1995). Additionally, radially lost oxygen around sedge roots can enhance nitrogen mineralization (Sun et al. 2019). Thus, high abundance of sedges in group 2 may increase nitrogen availability compared to the other groups. In groups 1 and 3 above to belowground ratio of sedge biomass is lower, compared to group 2, likely due to lower nutrient availability. Root morphology of sedges was not different between the groups, except for root diameter and surface area between group 1 and 2 (Appendix Table A 8). In group 2 surface area was higher, but specific surface area was similar, since also root biomass in group 2 was greater. In group 1 root diameter was smaller compared to group 2. Since post-thaw vegetation in group 1 was sparce and established only recently, less well-decomposable substrate such as sedge litter and lower oxygen supply

by roots may have led to low mineralization rates (Updegraff et al. 1995; Sun et al. 2019). Smaller root diameter can enhance nitrogen uptake (Hong et al. 2018; Freschet et al. 2021), which may explain smaller root diameter in group 1 compared to the other groups. Thus, variation in biomass and root morphology likely relate to successional stage and nutrient availability.

4.3.3 Methane emissions from sedge and non-sedge patches

Aligning with hypothesis 3.2 thermokarst methane emissions were 3.4 times higher in sedge compared to non-sedge patches in late summer (end of July to beginning of September). In these patches the only occurring sedge was *Eriophorum scheuchzeri*, with 20% of the total biomass (Appendix Table A 7). Both sedge and non-sedge patches were covered by the moss *Sphagnum riparium.* To prevent mutual influence between measurements in sedge and non-sedge patches, 30 cm deep collars were installed. Since collars were directly adjacent to each other similar temperatures and water table depth were obtained. High sedge affected methane emissions were also found in other thermokarst and non-permafrost affected mires (Ström et al. 2012; Korrensalo et al. 2022; Goud et al. 2022; Jentzsch et al. 2024; Mollenkopf et al. 2026). Here, mean methane emissions in sedge patches were 31 mmol m^{-2} d^{-1} in late summer and 21 mmol m^{-2} d^{-1} from the beginning of July to the beginning of September. These emissions are high compared to reported sedge affected methane emissions in literature. For example, sedge affected methane emissions in a young thermokarst with mean 5 mmol m^{-2} d^{-1} were measured from May to September (Heffernan et al. 2022). From end of June to the beginning of August mean 17 mmol CH_4 m^{-2} d^{-1} were measured in a thermokarst in Greenland (Ström et al. 2012). Measurements in an ombrotrophic bog in southern Finnland with 21 mmol m^{-2} d^{-1} were similar to methane emissions here measured in July and August respectively (Jentzsch et al. 2024). Ding et al. (2005) measured methane emissions in a marsh in northeast China exceeding those found here. There, mean methane emissions varied with 23 to 45 mmol m^{-2} d^{-1} between two dominating *Carex* species and two years measured in August (Ding et al. 2005). Taken together here methane emissions in sedge patches were relatively high compared to reported emissions in literature and higher compared to non-sedge patches.

4.3.4 Processes relating to methane emissions in sedge patches

Methane emissions are affected by sedges through several processes, based on the two main methane production pathways, the hydrogenotrophic and the acetoclastic pathway. In the hydrogenotrophic pathway CO_2 gets reduced to CH_4 and H_2 oxidized. In the acetoclastic pathway acetate gets split and the methyl group gets reduced. Acetate, CO_2, and H_2 are produced during the fermentation of peat and relatively new plant litter (Ferry 2011; Laanbroek 2010). Increased labile carbon supply through sedge litter and exudates, enhances production of acetate as a result of biomass fermentation. This in turn can enhance methane production from acetate, the acetoclastic methanogenesis and from CO_2, the hydrogenotrophic methanogenesis (Ferry 2011; Ström et al. 2012). Aerenchyma in sedges additionally allow bidirectional gas diffusion and can thus transport methane to the atmosphere. With plant mediated transport methane bypasses aerobic moss layers above the water table and with lower percentage methanotrophs living in symbiosis with *Sphagnum* mosses (Magnússon et al. 2020; Jentzsch et al. 2024; Mollenkopf et al. 2026). Thus, less methane gets oxidized during plant mediated transport and more methane is emitted. Simultaneously, during oxygen transport from the atmosphere to the roots, oxygen gets lost radially around roots (Vartapetian and Jackson 1997). This can enhance methane oxidization to CO_2 in the anaerobic layer and reduce methane emissions (Kankaala and Bergström 2004; Agethen et al. 2018). High labile carbon supply on the other hand can enhance use of oxygen and other electron acceptors and lower redox potential, which methane production is depended on (van Hulzen et al. 1999; Joabsson et al. 1999; Jentzsch et al. 2024; Mollenkopf et al. 2026). This highlights the role of substrate and occurrence of electron acceptors with higher redox potential for determining sedge effects on methane emissions (van Hulzen et al. 1999; Agethen et al. 2018). Gas transport capacity of plants can strongly vary among species and thus change relation of methane oxidized, and produced and emitted among species (Lai et al. 2012; Haviland and Noyce 2024). Variation in methane emissions among species was related to plant size, leaf area and resource allocation strategies, such as light use efficiency and foliar N, probably related to rooting depth (Goud et al. 2022). Additionally, there was more methane released at intermediately high compared to low abundance of sedges, due to a higher supply of labile C (Ström et al. 2012). Other studies showed lower methane transport capacities per biomass at high densities, as methane gets distributed more (Korrensalo et al. 2022; Jentzsch et al. 2024). Ström et al. (2012) compared high to low abundance of *Eriophorum scheuchzeri* and found double as high emissions in patches dominated by *E. scheuchzeri*. Here, *E. scheuchzeri* patches were compared to absence and not to lower abundance of sedges, which may at least partially explain higher difference of 3.4-fold increased emissions in sedge

compared to non-sedge patches. Overall, here and in other studies sedge induced methane production was greater than methane oxidation, which increased methane emissions with high sedge abundance. Studies have attributed increased methane emissions to plant mediated transport and labile carbon supply (Laanbroek 2010; Ström et al. 2012; Jentzsch et al. 2024; Mollenkopf et al. 2026). The percentual contributions of these processes and of the two main methane production pathways to the increased methane emissions in sedge patches at Ádejávri however remain unknown.

4.3.5 Seasonality of methane emissions

Methane emissions were highest in August in sedge patches, while at the beginning of the growing season mean emissions were lower and more similar in sedge and non-sedge patches. Here, soil temperature was not directly related to seasonally increased methane emissions, as soil temperature was lowest where fluxes were highest. Still, soil temperature is an important variable for methane emissions changing with season, since under water saturation higher soil temperature was found to enhance methane production and emissions (Rath et al. 2002; Jentzsch et al. 2024; Mollenkopf et al. 2026). Thus, soil temperature was likely related to methane emissions with delay through several processes. Higher soil temperatures accelerate electron acceptor cycling and more quickly allow methane production (van Hulzen et al. 1999). Additionally, temperature was found to increase growth rate of sedges (Korrensalo et al. 2022; Mollenkopf et al. 2026). Similar to here, others found highest methane emissions in August and June to August respectively, which was partly attributed to increased leaf area index and higher methane transport rates (Mollenkopf et al. 2026; Jentzsch et al. 2024). This fits well with highest sedge growth rates in June and July (Sloan et al. 2016). Some studies found that fast use of electron acceptors through higher labile carbon supply reduced relative abundance of methanotrophs as well as proportion of methane oxidation over season. Accordingly redox potential decreased with ongoing growing season, which explained the increase of methane emissions well (Turner et al. 2020; Mollenkopf et al. 2026). Consistent with these processes methane was found to increase in pore water over the growing season enhancing the efflux (Jentzsch et al. 2024). Thus, also here soil temperature likely increased methane emissions with a delay by accelerated electron acceptor cycling and plant growth. Plant growth and labile carbon supply may explain the greater increase in methane emissions with growing season in sedge compared to non-sedge patches.

The high abundance of sedges after permafrost collapse, has a high potential for increasing methane emissions during the growing season and may even contribute to a positive net trace gas radiative forcing for a period of time (Varner et al. 2022). Taking sedge abundance and seasonality into account is thus crucial for estimating methane emissions in thermokarsts.

5. Conclusion

Plant height influences the temperature regime within the peat body of relatively intact palsa formations. The shallower active layer beneath *Betula nana* (vegetation height 8 - 41 cm) compared to that beneath low vegetation (vegetation height <12 cm) suggests, that dense shrub cover could slow down the thawing of permafrost in patches on palsas. While *Betula nana* shrubs increase root and aboveground litter, it is unclear whether it acccumulates in the peat or whether it increases decompositon. Water uptake of *Betula nana* decreases soil moisture in summer, which decreases decomposition in already dry palsas.

Along the palsa slopes plant species composition did not differ, with the exception of the slope base. With the degradation and collapse of palsas, species composition shifted at the base of the palsa slopes, at the transition to thermokarsts. While plant diversity may decrease locally within peatlands with the full subsidence of palsas, regional plant diversity may not be affected, since the plant species found here, are not unique to palsa habitats. With the transition to thermokarsts water accumulated and formed new habitats with dominating *Sphagnum* and sedge species. Sedges can increase methane emissions during the growing season, while they also bind carbon dioxide through photosynthesis and peat accumulation. The net effect remains unclear and may differ between species and sites, and therefore needs further investigation.

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8. Appendix

Table A 1: Mean, maximum and minimum values from measured active layer depth and vegetation height in *B. nana* and low vegetation plots.

Measurements	***B. nana***	**Low vegetation**
Mean active layer depth (mean ± sd) (cm)	43.9 ± 5.07	53.4 ± 6.21
Min active layer depth (cm)	37.0	43.5
Max active layer depth (cm)	53.5	64.3
Mean vegetation height (mean ± sd) (cm)	25.3 ± 8.25	6.78 ± 1.72
Min vegetation height (cm)	8.00	4.70
Max vegetation height (cm)	41.3	12.2

Table A 2: Results from Kruskal-Wallis test and post-hoc Dunn test that show comparison between vegetation cover of species and groups between *B. nana* and low vegetation plots.

Plant species cover comparison *Betula nana* and low vegetation	**P-value**
Betula nana	**5.61 e-12**
Rhododendron tomentosum	**0.008**
Empetrum nigrum	0.227
Vaccinium vitis-idea	0.251
Vaccinium uliginosum	0.366
Andromeda polifolia	0.459
Rubus chamaemorus	0.288
Lichen	0.182
Bryophytes	0.312
Total Cover	0.362

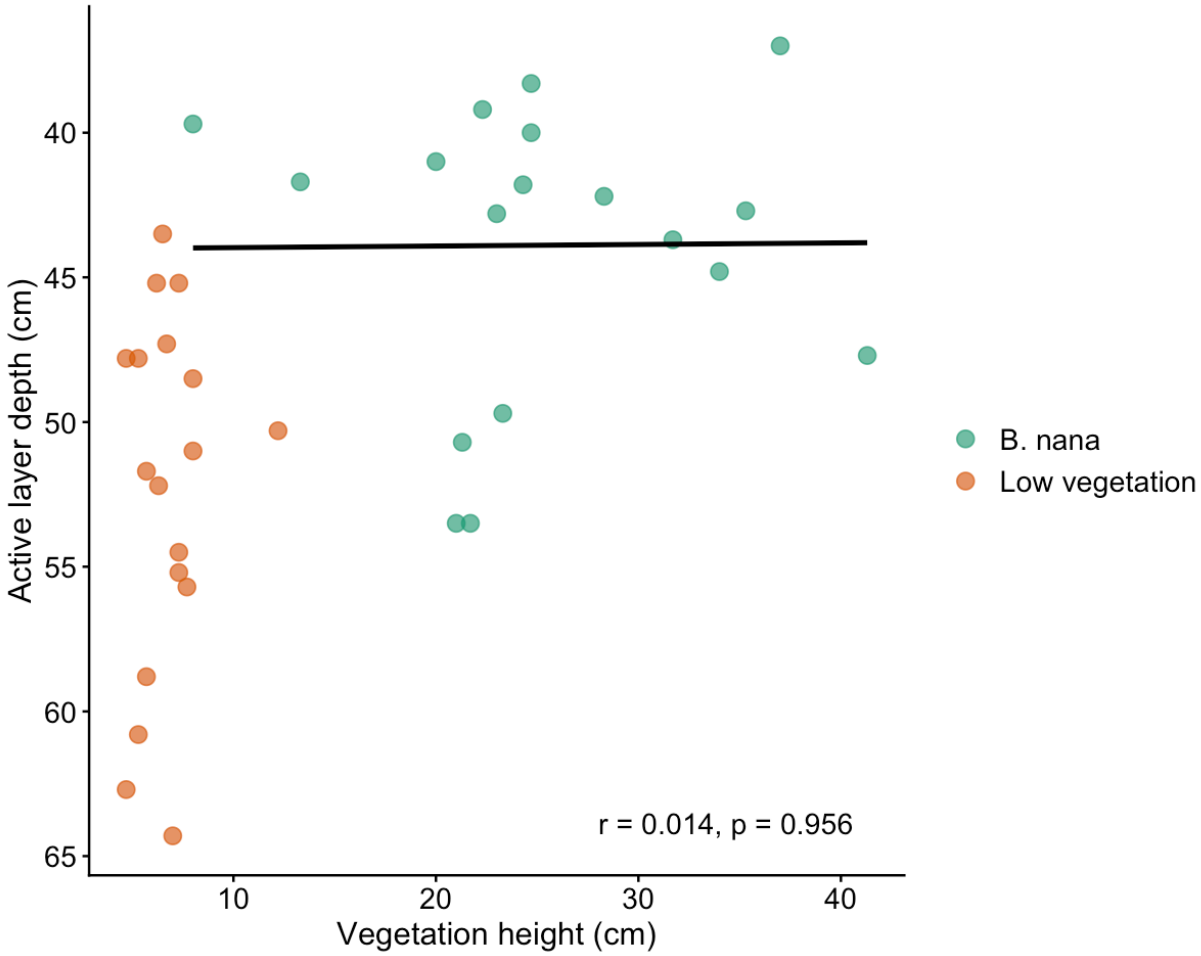


Figure A 1: Active layer depth (Y-axis, cm) against vegetation height (Y-axis, cm) with low vegetation in orange and *B. nana* in turquoise. Correlation and regression were done for *B. nana* only (pearson correlation).

Table A 3: PERMANOVA results showing differences in species composition with explanatory variables ‘slope section’ and ‘palsa slope’.

Model variables	Degrees of freedom	Sum of squares	R^2	F	P
Slope section	3	0.63	0.14	3.78	**0.001**
Palsa slope	9	1.11	0.25	2.20	**0.001**
Model	12	1.80	0.40	2.68	**0.001**
Residual	48	2.69	0.60		
Total	60	4.49	1		

Table A 4: Results from pairwise PERMANOVA shows difference in species composition across slope sections palsa top, upper slope, lower slope and slope base.

Slope section	Palsa top	Upper slope	Lower slope
Upper slope	p = 0.310	-	-
Lower slope	p = 0.664	p = 0.664	-
Slope base	**p = 0.006**	**p = 0.012**	**p = 0.016**

Table A 5: Mean, minimum and maximum values of active layer depth along the palsa slopes from palsa top to slope base. Active layer depth >160 cm was not measured.

Active layer depth	Palsa top	Upper slope	Lower slope	Slope base
Mean (cm)*	64	85	70	120
Min (cm)	45	44	43	85
Max (cm)	122	>160	>160	>160

* Excluding >160cm

Table A 6: Shows mean relative abundance for species at every slope section and significant codes for species that contributed significantly to difference in species composition between groups (see letters). *Trientalis europaea*, *Arctostaphylos alpina*, *Vaccinium oxycoccos* and *Betula pubescens* were not included in later analysis.

Species, Total cover	Palsa top (T), species cover %	Upper slope (U) species cover %	Lower slope (L) species cover %	Slope base (B) species cover %
Betula nana	1.8	7.7	5.4	7.6
Betula pubescens	0	0	0	< 0.1
Vaccinium uliginosum (E.)	1.1	1.9	1.4	1.8
Vaccinium myrtillus (E.)	0	0.6	1.0	3.0
Arctostaphylos alpina (E.)	0.3	0	0	0
Rhododendron tomentosum (E.)	2.3	4.4	2.6	2.8
Empetrum nigrum (E.)	51.8 (B)*	39.6	33.7	11.6 (T)*
Vaccinium vitis-idaea (E.)	3.9	3.2	2.4	1.6
Vaccinium oxycoccos (E.)	0	0	0	0.3
Andromeda polifolia (E.)	0.9	2.3	1.6	1.4
Rubus chamaemorus	10.8 (B)***	7.5 (B)**	9.3 (B)**	25.7 (T)*** (U, L)**
Trientalis europaea	0	0	0	< 0.1
Carex glacialis	0	0	0	0.1
Carex rariflora	0	0	0	0.1
Carex rotundata	0	0	0	1.3
Carex rostrata	0	0	0	1.1
Eriophorum angustifolium	0	0	0	< 0.1
Eriophorum russeolum	0	0	0	3.2
Eriophorum vaginatum	0	0	0	0.2
Sedge sum	0 (B)***	0	0	5.5 (T)***
Bryophyte	17.7 (B).	17.9 (B).	28.2 (B).	31.9 (T, U, L).
Lichen	9.4	14.9	14.3	6.2
Total cover	82.5 (B)*	93.5	85.0	95.8 (T)*

T stands for the palsa top, U for the upper slope, L for the lower slope, B for the slope base.

Significant codes are 0 '***', 0.001 '**', 0.01 '*' 0.05 '.'.

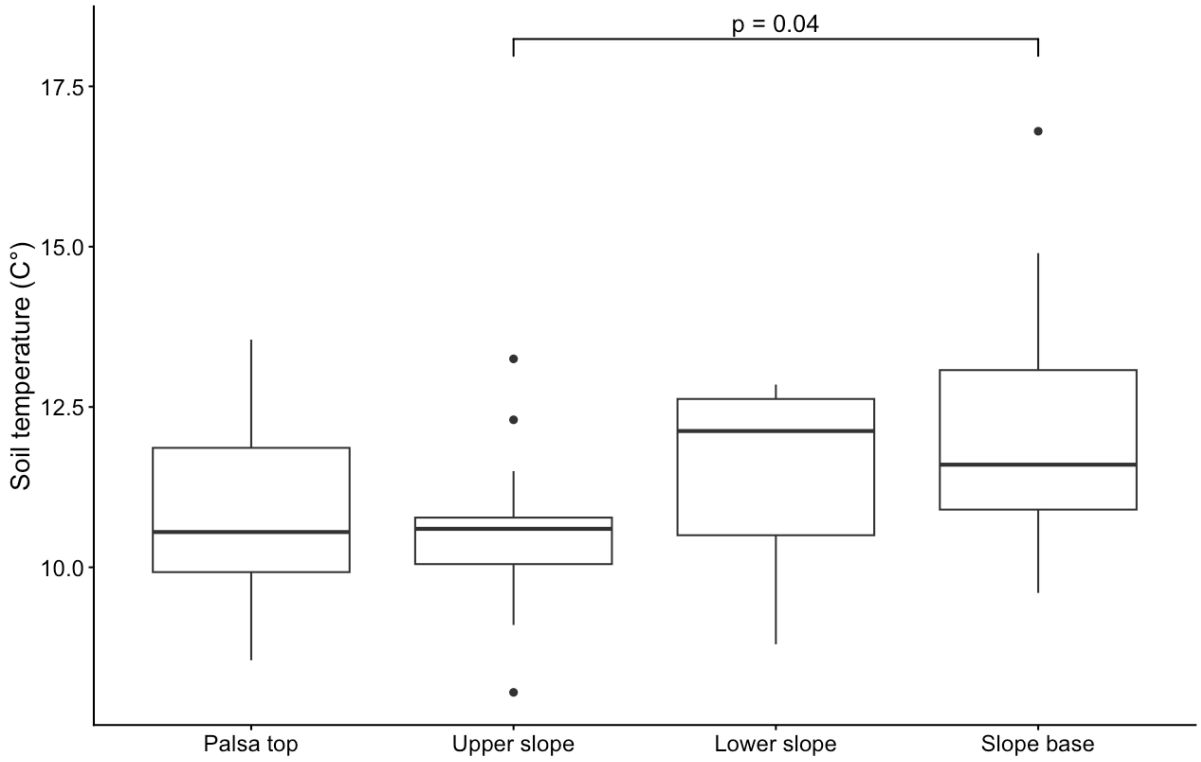


Figure A 2: Soil temperature (C°) in the slope sections from palsa top to slope base. Boxplots show median (centre line), 0.25 and 0.75 percentile (lower and upper margin) of soil temperature. Probability was derived from ANOVA, showing only a difference in soil temperature between upper slope and slope base.

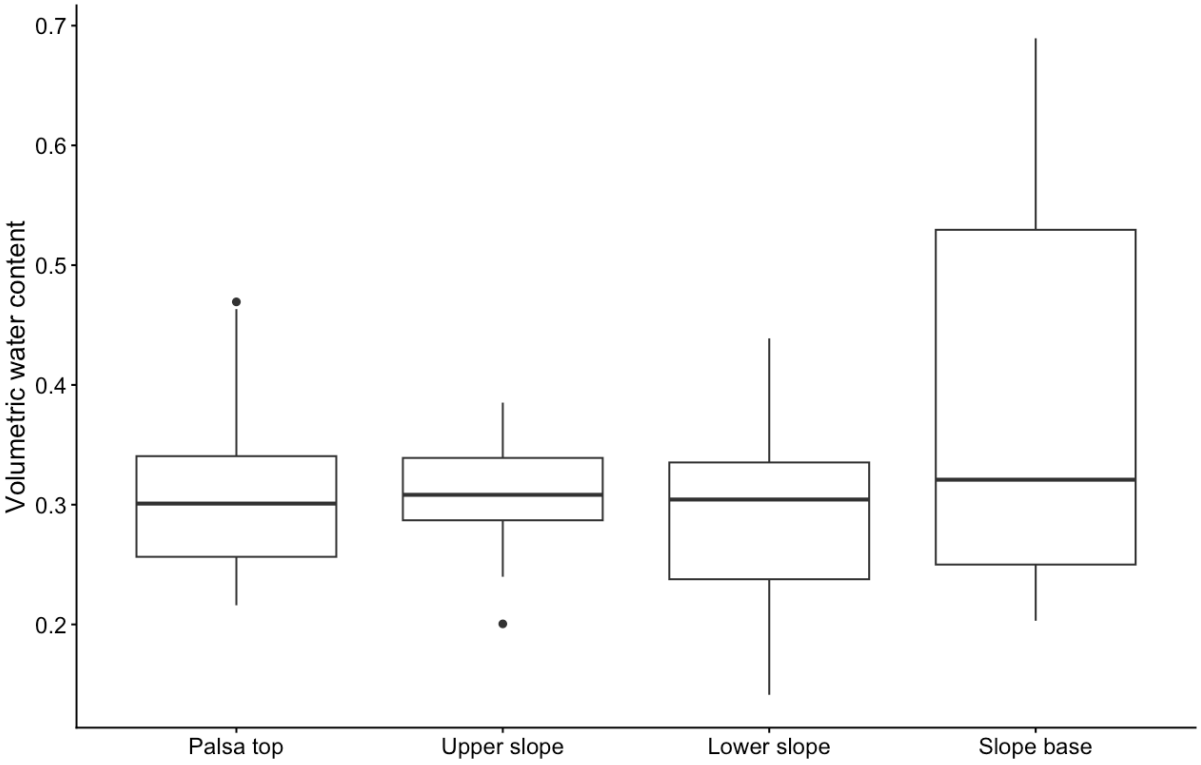


Figure A 3: Volumetric water content in the slope sections from palsa top to slope base. Boxplots show median (centre line), 0.25 and 0.75 percentile (lower and upper margin) of soil volumetric water content. ANOVA for deriving difference in volumetric water content between slope sections was done, however, sections did not differ in volumetric water content.

Table A 7: Biomass (mean ± sd) of three vegetation zones AT1, AT3 and AT5 at a thermokarst in Ádejávri belonging to the vegetation composition groups 1, 2 and 3 respectively. P-value were derived from ANOVA comparison. Total biomass and above ground biomass were square root transformed and root mass and *Sphagnum* weight was logarithm transformed to meet the assumption of a normal distribution and homoscedasticity.

Biomass	**AT1**	**AT3**	**AT5**	**AT3 – AT1**	**AT5 – AT1**	**AT5 – AT3**
Total (g/m^2)	6.79 ± 3.62	$2.18 \cdot 10^2 \pm 0.40 \cdot 10^2$	$2.46 \cdot 10^2 \pm 42.74$	**p < 0.001**	**p < 0.001**	p = 0.49
Above Ground (g/m^2)	5.47 ± 3.04	$2.14 \cdot 10^2 \pm 39.52$	$2.44 \cdot 10^2 \pm 42.54$	**p < 0.001**	**p < 0.001**	p = 0.43
Sedge (g/m^2)	6.79 ± 3.62	45.7 ± 6.62	13.34 ± 2.4	**p < 0.001**	p = 0.10	**p < 0.001**
Sedge aboveground (g/m^2)	5.47 ± 3.04	41.2 ± 4.96	10.96 ± 1.95	**p < 0.001**	p = 0.07	**p < 0.001**
Sedge roots (g/m^2)	1.31 ± 0.63	4.48 ± 2.98	2.38 ± 0.74	**p = 0.006**	p = 0.12	p = 0.25
Ratio above-belowground	4.13 ± 1.32	9.20 ± 5.34	4.87 ± 1.22	**p = 0.005**	p = 0.86	**p = 0.01**
Sphagnum (g/m^2)	0 ± 0	$1.73 \cdot 10^2 \pm 37.7$	$2.33 \cdot 10^2 \pm 43.1$	/	/	p = 0.05

Table A 8: Shows root morphological traits diameter (mm), length (m), specific length (m/g), surface area (cm^2) and specific surface area (m^2/g) (mean ± sd) of sedges at vegetation zones AT1, AT3, AT5 belonging to the vegetation composition groups 1, 2 and 3 respectively at a thermokarst in Ádejávri. Root length, surface area, and specific surface area were log-transformed into a normal distribution in order to use ANOVA to determine p-values for differences in root morphology between vegetation zones.

Root morphological parameter	**AT1**	**AT3**	**AT5**	**AT3 – AT1**	**AT5 – AT1**	**AT5 – AT3**
Diameter (mm)	0.53 ± 0.05	0.7 ± 0.08	0.61 ± 0.05	**p = 0.003**	p = 0.12	p = 0.13
Length (m)	1.97 ± 0.55	5.04 ± 3.79	2.77 ± 1.87	p = 0.097	p = 0.80	p = 0.27
Specific Length (m/g)	62.8 ± 27.2	40.7 ± 12.7	42.1 ± 19.7	p = 0.25	p = 0.29	p = 0.99
Surface Area (cm^2)	32.5 ± 8.84	$1.08 \cdot 10^2$ ± 79.4	53.2 ± 34.1	**p = 0.02**	p = 0.53	p = 0.14
Specific Surface Area (m^2/g)	10.1 ± 3.42	8.7 ± 1.73	8.15 ± 3.83	p = 0.93	p = 0.60	p = 0.81